

Chapter 1

Question III: Head-Related Transfer Functions (HRTFs) and Machine Learning

1.1 Introduction

What is Head-Related Transfer Function (HRTF)?

Sound localization is a critical auditory function that enables humans and animals to determine the origin of sounds in their environment. This capability is fundamental to survival and has significant evolutionary importance, aiding in locating prey, evading predators, and navigating social interactions. In social animals, including humans, this ability facilitates communication by enabling individuals to identify the source of sounds, understand spatial arrangements and navigate complex environments. The process of sound localization involves intricate physicoacoustic and neurological mechanisms, which have evolved to allow organisms to accurately interpret acoustic information.

Humans localize sound using several auditory cues, which can be broadly classified into binaural (involving both ears) and monaural (involving one ear) cues. Binaural Cues are Interaural Time Differences (ITD) and interaural level differences (ILD) that help the human brain identify the direction of sound sources. The brain analyzes these time and level differences to localize sound sources. Additionally, the monaural cues give some fine-tuned information to the brain. These cues depend on the acoustic properties that can be perceived by a single ear. The most important among these is the sound's spectral composition, which is shaped by the unique configuration of the ear's anatomy, the specifics of acoustic shadows cast by the individual's head, and reflections from the individual's body. These person-specific differences result in different frequencies being absorbed or reflected differently from different positions, providing the brain with nuanced information about the sound's location. Such cues are especially valuable for pinpointing the vertical position of sounds and discerning whether sounds originate from in front or behind the listener. The brain also utilizes other cues as additional information, such as distance and dynamic cues. These include the overall level of the sound, the ratio of direct to reverberant sound energy, frequency content (with high frequencies being attenuated more over distance than low frequencies), and variations in loudness, pitch, and spectral content of the sound for tracking moving objects.

Head-Related Transfer Functions (HRTFs) are mathematical functions used to describe how ears receive a sound from a point in space. HRTFs encapsulate ITD, ILD, spectral cues and all the complex interplay of factors influencing how sound is perceived by an individual's ears from any given point in space. Essentially, HRTFs represent the mathematical modifications sound waves undergo while traveling from the the given position to human's ears, with its interaction with the human body, including reflections and diffraction around the head, ears, and torso, before finally reaching the ear canal. By applying HRTFs, audio systems, such as headphones, can simulate how sounds would naturally arrive at a listener's ears from different directions, enhancing the perception of spatial location and distance.

HRTFs are critical in the realm of media technology for auditory perception, particularly within the scope of virtual reality (VR), augmented reality (AR), video games, multimedia systems, binaural audio for media, hearing aids, and telecommunication systems where spatial audio enhances the user experience and other spatial audio renderings. In multimedia and gaming, HRTFs contribute to a more engaging and realistic environment, allowing users to perceive audio cues with directional accuracy. For hearing aids, HRTFs can improve spatial awareness and localization, essential for navigating complex auditory scenes.

Beyond these applications, the influence of HRTFs extends into the broader spheres of art and media, where they revolutionize the way audiences experience sound. In cinematic productions and music, HRTFs allow for the creation of deeply immersive soundscapes that envelop the audience, transcending traditional stereo techniques to offer a more nuanced and spatially aware hearing experience. Spatial audio through HRTF enables artists and creators to manipulate sound in three-dimensional space, delivering a more engaging and emotionally resonant connection with their audiences. In performances and installations, HRTFs facilitate the design of complex acoustic environments that can mimic real-world soundscapes or create entirely new auditory experiences, challenging the audience's perception and enhancing artistic expression. As such, HRTFs are not merely technical tools but also artistic instruments that expand the palette of possibilities for creators, allowing them

to explore and convey narratives and emotions through the medium of sound.

1.2 Background

1.2.1 Localization of Sound

Being binaural, human hearing encompasses not only the perceptions of loudness, pitch, and timbre but also includes a critical aspect known as spatial perception. This perception allows humans to evaluate the spatial position of sound sources, including their direction and distance, aiding in visual attention and serving as a warning mechanism for potential dangers or a general orienting mechanism. Studies have shown that the ability to locate a sound source is influenced by various factors, such as the sound's direction, distance, and the acoustical properties of an individual's body, head, and ears. These factors vary among individuals, and therefore, the mechanisms that human brains use to decode spatial information are unique for every person.

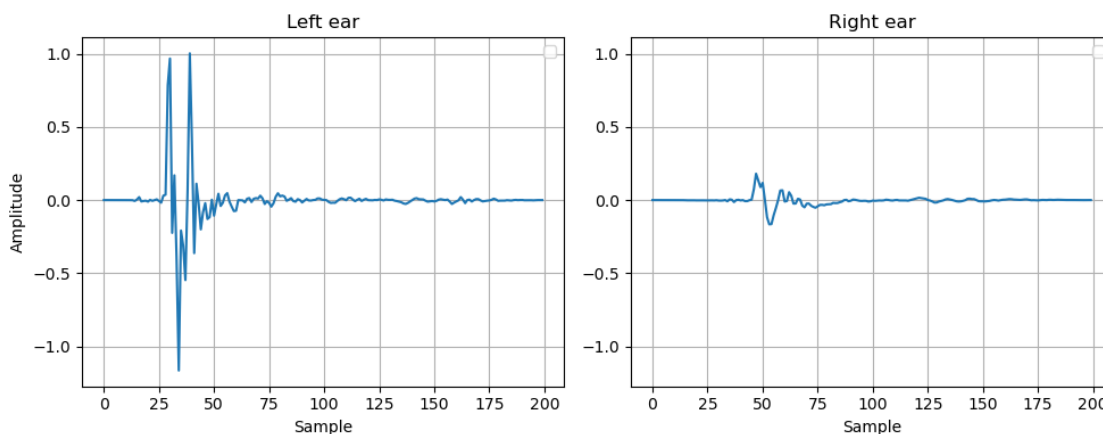


Figure 1.2.1: Time and volume differences for sound coming from the left side at 45 degrees with an elevation of 0 from 1 meter distance. The image is plotted from the CIPIC dataset.

Psychoacoustic research has identified several key cues that humans use for directional localization of a single sound source. These cues include the interaural time difference (ITD), where the timing difference in sound arrival between the two ears helps determine the direction of the sound source; the interaural level difference (ILD), which refers to the

difference in sound pressure level reaching each ear, aiding in localization; as well spectral cues that pertain to the frequency content of the sound. Sounds originating from one side of the body will reach the closer ear slightly sooner than the farther ear. The brain uses these timing differences to determine the sound's direction. Also, due to the slightly longer distance and head's shadow effect, a sound from one side will be slightly quieter in the ear further away. These elements work together to provide an understanding of where a sound is coming from. Figure 1.2.1 illustrates the waveform of the same sound captured by microphones positioned at the left and right ears, showcasing the differences in between positions. This recording simulates the auditory perspective of the left and right ears, with the sound originating from a position 45 degrees to the left of the head at ear level (no elevation) and from a distance of 1 meter. As we can see, ITD and ILD are clearly identifiable on the figure. The waveform shape clearly indicates that the sound is louder on the left side due to the fact that the sound source is closer to the left ear than the right ear, resulting in a stronger sound intensity. Additionally, the sound arrives faster at the left ear compared to the right ear due to the shorter distance the sound wave travels from the source to the left ear, resulting in a shorter time delay.

Furthermore, the perception of distance is informed by a comprehensive set of cues, not just limited to ITD, ILD, but also including factors such as the spectral cues, relative intensity of the sound, the ratio of direct to reverberant sound energy, and changes in the spectral content due to air absorption over distance. Complex interplay of all these factors allow individuals to not only perceive the direction but also estimate the distance of sound sources in their environment, showcasing the auditory system's remarkable capacity to interpret complex acoustic information. Let us examine these key factors in more details.

Interaural Time Difference

The Interaural Time Difference (ITD) plays a crucial role in the localization of sound, referring to the difference in sound arrival times at the two ears. This concept was described

in Woodworth and Schlosberg's study [27] and can be encapsulated mathematically as:

$$ITD = \frac{R}{c} (\theta + \sin(\theta)) \quad (1.2.1)$$

Where, R represents the head's radius, θ denotes the sound wave's angle of incidence relative to the head's center line, measured in radians, and c is the speed of sound in air. The ITD value is zero when the sound source is directly in front (0°) or behind (180°) of the listener, reaching a maximum when the sound comes from the sides (90° and -90°), due to the differential sound wave arrival times at each ear. This is illustrated in Figure 1.2.2.

ITD is particularly relevant for frequencies where the sound's wavelength is comparable to the head's size. At wavelengths shorter than the head's width, the head creates an acoustic shadow for higher frequencies, whereas for lower frequencies, sound waves can diffract and travel around the head.

Other than frequency ITDs are also significantly influenced by the anatomical dimensions of an individual's head, which vary from person to person. Consequently, ITD experiences are inherently individual-specific. The complexity introduced by different frequencies behaving differently and individual variability means that establishing a universally applicable, simple formula for calculating ITD is challenging.

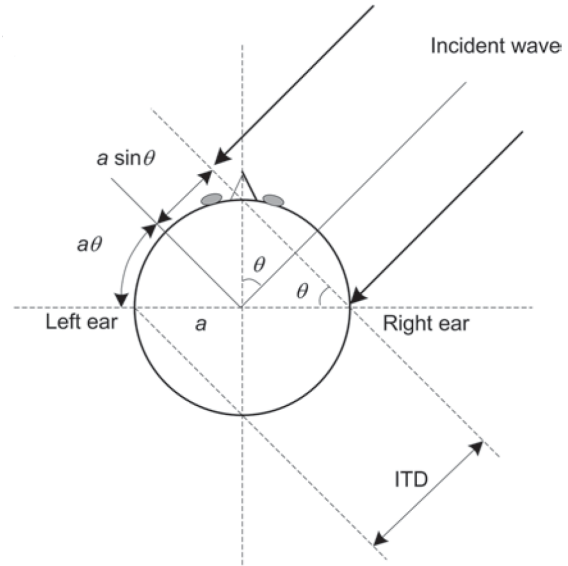


Figure 1.2.2: Illustration of ITD calculation considering sound movement around the head.

Interaural Level Difference

Interaural Level Difference (ILD) is another important cue to directional localization. When a sound source deviates from the certain directions, the sound pressure at the farther

ear is attenuated. Shown in Figure 1.2.3, the head creates an acoustic shadow which results in a reduction of the sound intensity at the ear farther away from the sound source. This difference is known as the ILD.

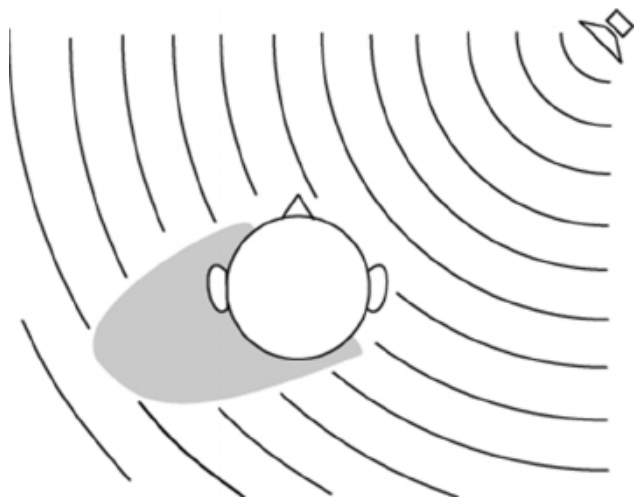


Figure 1.2.3: A graphical representation of the head shadow effect causing ILD.

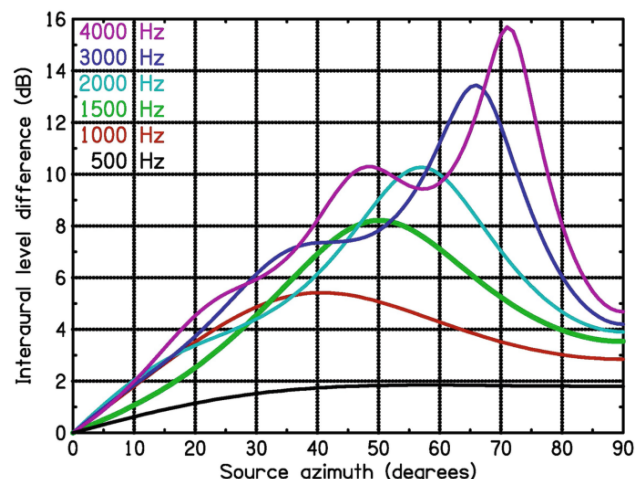


Figure 1.2.4: Illustration of ILD for different frequencies and azimuth angles

ILD is especially important at high frequencies because high frequencies are shadowed by the head while low frequencies can travel around it. This acoustic phenomenon leads to direction- and frequency-dependent ILDs. Figure 1.2.4 depicts the complex relationship between the position of a sound source and the ILD across various frequencies. At lower frequencies, the ILD changes gradually, with the maximum ILD differing by only 2 dB from the central position where ILD is zero to the 90° point, while higher frequencies show drastic changes up to 16 dB.

Additionally, Contrary to what might be expected, the figure demonstrates that the peak ILD for higher frequency sinusoidal sounds (for complex sounds that comprise multiple frequency components the ILD changes more uniformly with azimuth) does not occur at an azimuth of 90°, where one ear is entirely in the acoustic shadow.

This phenomenon is attributed to the constructive interference of sound waves diffracting around the head, which can amplify the sound pressure at the ear opposite the sound source.

This demonstrates that ILD is a function influenced by multiple variables, making it

dependent on both the direction and frequency relative to a specific head radius. Similar to ITD, ILD is influenced by the anatomical dimensions of the individual, facial characteristics, pinnae and other intricate structures, meaning that it is also a personalized cue for localization.

Spectral Cues

The ITD and ILD are fundamental acoustic cues used for sound localization, particularly in the horizontal plane. While they are strong predictors in discerning the location of sounds at low and high frequencies, respectively [7], these cues alone are not sufficient for precisely pinpointing the unique position of a sound source in three-dimensional space.

A hypothetical scenario where the head is considered a perfect sphere and ears are two points, gives rise to what is known as the "cone of confusion" problem. On this cone, depicted on the figure 1.2.5, theoretically infinite spatial positions yield the same ITD and ILD, making it impossible to locate a sound source based solely on this cue. For sounds coming from this cone the front-back

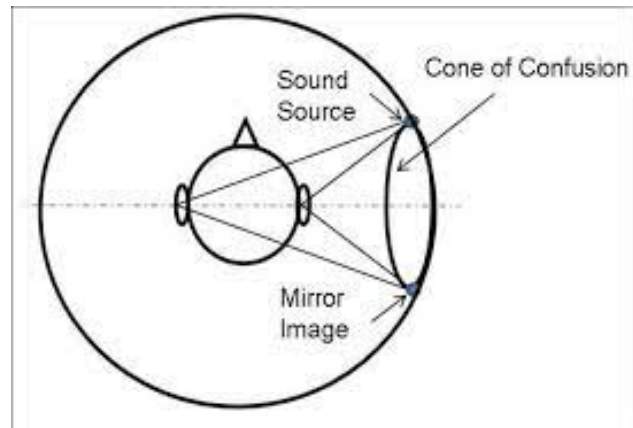


Figure 1.2.5: Illustration of cone of confusion

disambiguation becomes difficult. For instance, for sound sources at azimuths of 45° (right front) and 135° (right back) the ITD and ILD are identical, with a symmetrical spherical head model. "Cone of confusion" is not only a problem for idealized mathematical model. Complications also arise in practical scenarios, even though the human head is not a perfect sphere and possesses a more complex geometry. Here, the ITD and ILD still fail to offer a straightforward correlation with sound source positions, and thus, the cone of confusion persists, albeit not as a perfect geometric shape. Another particularly challenging scenario for these localization cues are the conditions on the median plane, which is a special case of

”cone on confusion,” where ITD and ILD cues are nearly or exactly zero due to the symmetrical sound pressure on both ears. This gives rise to so called front-back confusions. Another problem for ILD is in a far-field situation, where sound distance greatly exceeds head radius, different points may produce identical ILDs.

Since ITD and ILD are not fully capable of identifying the exact spatial location, especially in 3-dimensional space, additional cues—particularly spectral cues shaped by the pinna, head, and torso reflections and diffractions—are crucial for vertical sound localization and resolving front-back confusions. These spectral cues, which are distinct from binaural cues like ITD and ILD, are monaural and thus pertain to the sound processing of each ear individually [26].

Spectral cues, created by the diffraction and reflection around anatomical structures, act as personalized auditory localization indicators. Establishing a precise and quantitative relationship between these spectral features and the sound source direction is challenging and can only be empirically accounted. Nevertheless, it is evident that individuals rely on their unique spectral cues to accurately identify sound source locations.

1.2.2 Head-Related Transfer Functions (HRTFs)

As we see ITD, ILD, and spectral cues are all highly complex, nonlinear and individual factors. The sound emitted from a point source travels to ears, undergoes different types of attenuation because of distance and position, interacts with anatomical structures, undergoing diffraction and reflection off the head, torso, and pinnae, before reaching both ears. This interaction results in binaural sound pressure change functions that carry their key localization cues information. The human brain then reads and integrates these cues to understand the sound source’s location. To measure and describe this complex process, head-related transfer functions (HRTFs) are used. HRTFs encapsulate the filtering effects of ITD, ILD and anatomical structures. For any given source position, the mathematical formulation of a pair of HRTFs, one for the left ear H_L and one for the right ear H_R , is characterizing the

acoustic transfer functions as follows:

$$\begin{aligned} H_L &= H_L(f, \varphi, \theta, a) = \frac{P_L(r, f, \varphi, \theta, a)}{P_0(r, f)}, \\ H_R &= H_R(f, \varphi, \theta, a) = \frac{P_R(r, f, \varphi, \theta, a)}{P_0(r, f)}, \end{aligned} \tag{1.2.2}$$

where P_L and P_R are the sound pressure functions in the frequency domain at the left and right ears, respectively. These pressures are functions of distance r , frequency f , azimuth φ , elevation θ angles, and a head-related parameter a . P_0 is a reference sound pressure function at a given frequency f and distance r , which would occur at the center of the head were the head absent. The HRTFs, H_L and H_R , are defined as transfer functions from reference sound pressure at a given frequency f and distance r to sound pressure (P_L and P_R) at the left and right ears, respectively. The distance r can often be disregarded in the formula, especially when examining the far-field HRTF, as it is largely distance-independent.

HRTFs are typically expressed in the frequency domain, showing how different frequencies are amplified or attenuated by distances and the anatomy of the head, ears, and body. The time domain counterpart of HRTFs and the most widely used method for measuring and storing HRTF information are Head-Related Impulse Responses (HRIRs). HRIRs essentially serve as time domain FIR filters for each ear and position and can be represented as audio recordings of an impulse sound from a given position, recorded at the position of the ear canal. These recordings contain information about all the transformations the sound wave undergoes before reaching the ear canal. By selecting a pair of left and right HRIRs for a specific location and applying the filter to the given sound, we can create an auditory sensation for the listener that the sound is emanating from the selected spot.

HRTFs and HRIRs typically capture all the information about what happens with sound from its source to the ear. However, in some practical cases ITD and ILD — the differences in timing and intensity of sound between the two ears — are calculated using traditional signal processing algorithms. These methods are preferred because they are relatively easier to implement. Meanwhile, the remaining spectral cues, which have complex dependencies

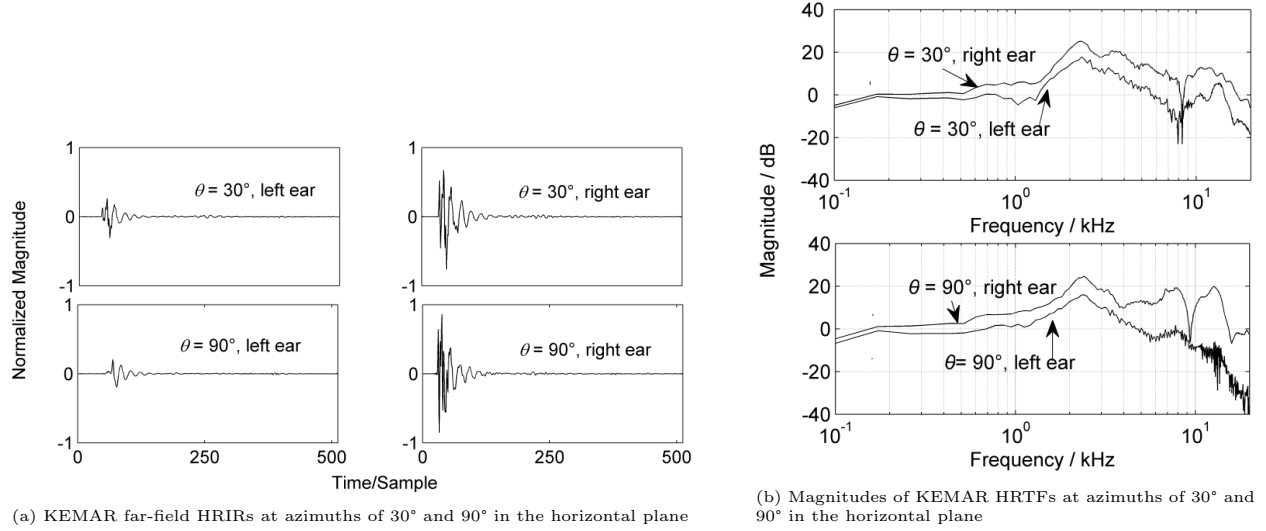


Figure 1.2.6: HRIRs and HRTFs for KEMAR model. Figure taken from [19]

on a person's anthropometric data, are incorporated into HRTFs. These HRTFs complement interaural cues by supplying the critical spectral information necessary for the brain to interpret spatial characteristics. This integration enables the precise localization of sound sources in three-dimensional space, allowing individuals to accurately determine the direction, elevation, and distance of sounds with remarkable precision.

The example of HRIR measurements and the corresponding HRTFs are shown in Figure 1.2.6, which illustrates the differences between the left and right ears in both the time domain and frequency domain for sounds coming from two different positions measured for the KEMAR mannequin. In subfigure (a), the far-field HRIRs at azimuths of 30° and 90° in the horizontal plane are presented for both the left and right ears. The normalized magnitude of the impulse response is plotted against time/sample. It can be observed that the impulse responses vary between the left and right ears as well as between the two azimuth angles, demonstrating the spatial characteristics captured by the HRIRs. Subfigure (b) shows the magnitudes of the corresponding HRTFs at the same azimuths of 30° and 90° . The magnitude in dB is plotted against the frequency in kHz. The plots highlight the frequency-dependent differences between the left and right ears for the given sound source positions. The variations in the HRTFs indicate how the ear shapes and head geometry influence the

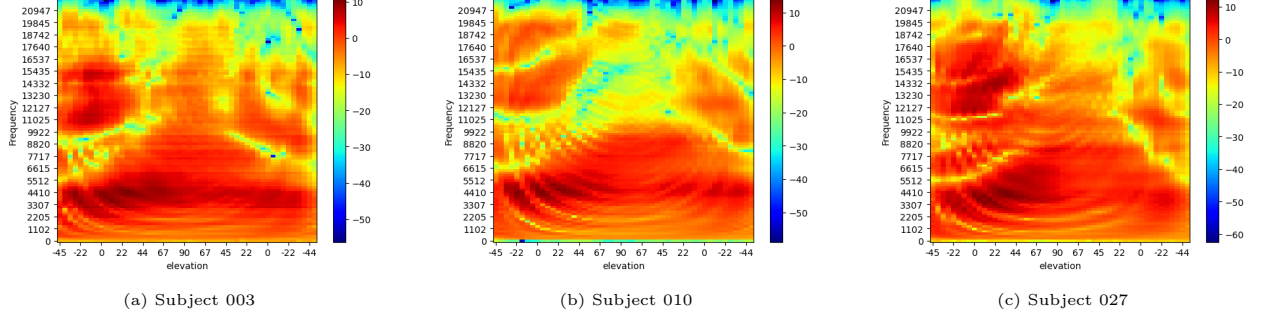


Figure 1.2.7: HRTF spectral images from the CIPIC database for the right ear for all elevations in azimuth angle 0.

frequency response, contributing to the spatial hearing cues necessary for accurate sound localization.

The HRTF varies from person to person due to the unique anatomical structures and dimensions of each individual, which are denoted by the variable a in Eq. 1.2.2. Specifically, a represents a comprehensive set of anatomical parameters that describe the dimensions of relevant bodily features. These differences are particularly pronounced in the median plane, where ITD and ILD are virtually zero. For example, Figure 1.2.7 demonstrates HRTFs' spectral images for all elevations at an azimuth of 0° for the right ear in the CIPIC database across several different subjects. As we see, the magnitudes of HRTFs significantly differ among different individuals even when observing the same directions.

1.2.3 HRTF and ML: Problem Statement and Key Challenges

Recently, with the advance of technology and the introduction of machine learning (ML), the question of modeling HRTFs with ML has emerged. As we have seen, HRTFs describe highly complex, nonlinear factors with numerous variables and nonlinear relationships, necessitating personalized approaches that are very hard to model with traditional mathematical and signal processing methods. ML is known for its ability to tackle such problems. The growing availability of HRTF datasets in the public domain, combined with the intuitive choice to address such a complex and nonlinear challenge through ML, has initiated a substantial increase in ML research efforts in the field. This wave has led scholars to investigate a

variety of input data formats and deploy a broad range of computational models.

The main objective of using ML to model HRTFs for human users is to solve personalization challenges and capture all the nuances necessary to provide users with realistic sound images. Ideally, ML could achieve model HRTF from accessible data, thereby avoiding the costly and time-consuming data collection procedures required for traditional HRTF acquisition. The ultimate goal would be for ML to generate HRTFs directly from anthropometric measurements, photos of the ear and torso, partially measured HRTFs and other similar inputs.

The field of spatial sound and HRTF modeling with ML has encountered several unresolved important challenges. These challenges both technical and conceptual, stemming from both the logistical complexities, a limited understanding of the underlying mechanisms and lack of sufficient methods and theory. The challenges in HRTF research can be categorised into four key areas:

Measurement Challenges: Collecting accurate HRTF data is a costly and invasive process. The requirement for dedicated facilities, such as anechoic chambers, and specialized audio equipment, coupled with the intrusive nature of the measurement procedures, renders the process impractical, especially for mass individualized data collection intended for a broad consumer base. Additionally, technical issues, such as head movement during data collection, can introduce inaccuracies. All existing measurements and datasets potentially contain these minor inaccuracies, complicating their utilization for individual or research purposes. This will be discussed in more detail in a later section (see: 1.3.1).

Furthermore, even when measurements are conducted in a laboratory setting with optimal accuracy, it is technically impossible to make measurements at continuous, infinite locations around the subject’s head by positioning speakers at all possible positions. This leads to the particular problem of interpolation and extrapolation in HRTF measurements. There has been a lot of work in this area, with varying degrees of success, and this problem is considered relatively well-solved today. More specific examples will be presented in the later

sections where we introduce existing works in the area (see: 1.4.2).

Scarcity of Datasets and Overfitting: ML models generally are data-intensive. However, existing datasets in HRTFs do not adequately meet these needs due to their limited size. Most datasets are too small, especially for deep learning models, which require datasets several magnitudes larger than what is currently available in the HRTF domain. This size discrepancy significantly increases the risk of overfitting.

Additionally, most datasets are compromised by inaccuracies. Many datasets suffer from problems related to the environments in which they are recorded; they are often collected in non-anechoic conditions, leading to contamination by echoes and standing waves that are unrelated to the HRTFs. This environmental noise complicates the learning process for models, making it difficult to discern the necessary patterns.

Understanding of Personalization: The evident necessity for personalization, essential for delivering a realistic and immersive auditory experience, requires that users have their own custom set of HRTFs. As previously discussed, conducting individual measurements is both expensive and complicated. This situation calls for the development of methods that can generate the necessary HRTF measurements using data that is easily and freely available, such as human anthropometric data, ear, and head photos, etc. However, to date, there are no effective ways of achieving this, and the issue of personalization remains largely unresolved.

Unfortunately, the precise mechanisms behind personalization, or at least the identification of the necessary data and information required for effective customization, still remain unclear and poorly understood. While numerous prediction models have been proposed, the representational power of various anthropometric input data is not fully clear. These models often use ear, torso, and head data that are selected empirically and labeled manually, yet there is no compelling evidence to date that they can predict HRTFs with high accuracy. It is not sufficiently studied and understood how clothing, the physical characteristics of the body surface and many other parameters of human body influence HRTFs. Using 2D ear images

for HRTF prediction, that has been often employed in studies recently, has its limitations, as the color images are often constrained by viewing angles and issues of self-occlusion.

Physical and Acoustic Accuracy: Ensuring that synthesized or interpolated HRTFs adhere to physical and acoustic principles is paramount. However, achieving this level of accuracy is challenging due to the limitations inherent in current data-driven modeling techniques. There have been a few interesting developments in the field attempting to incorporate principles of physics and acoustics into ML models, which will be discussed in detail later in the section on the overview of existing literature and work (see section: 1.4.3).

1.3 HRTF and HRIR Measurement, Datasets and Evaluation

1.3.1 Measuring HRTFs and HRIRs

HRTF measurements are complex and high costly. They are typically conducted in an anechoic chamber to simulate a free-field environment without walls and other reflective surfaces. In a basic HRTF acquisition setup utilizing digital measurement techniques, a subject is equipped with in-ear microphones, and a sound source, usually a high quality loudspeaker, is positioned at a specific location relative to the subject. An excitation signal generated by a computer is reproduced via the loudspeaker after passing through a digital-to-analog converter (DAC) and a power amplifier. The emitted signals are captured by the in-ear microphones, amplified, converted into a digital form through an analog-to-digital converter (ADC), and delivered back to the computer.

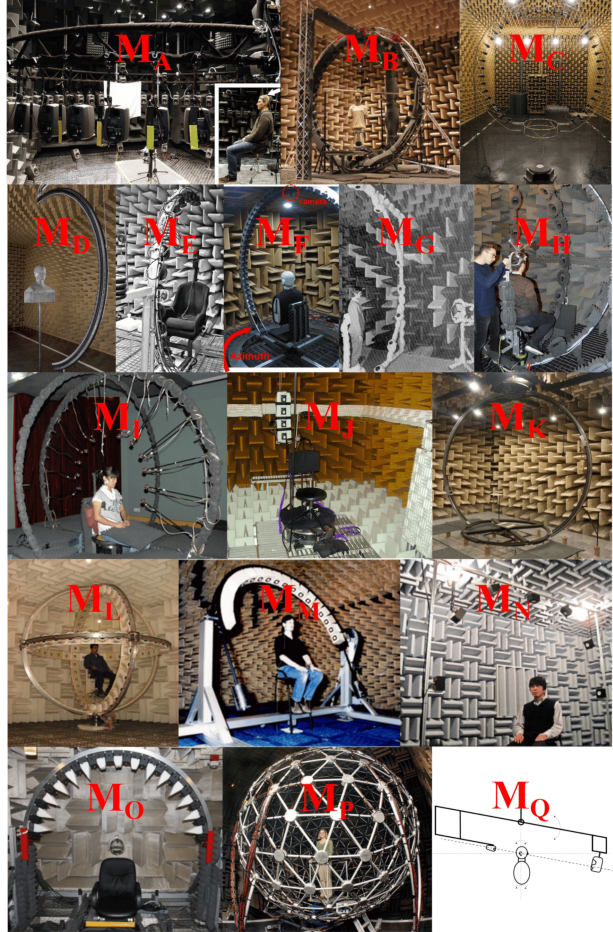


Figure 1.3.1: Some examples of multi-loudspeaker-based HRTF measurement setups. Figure taken from [18].

The recorded ear signals and the excitation signal are then used to calculate the pair of HRTFs. Figure 1.3.1 illustrates a few different real-world setups for measuring HRTFs.

To measure HRTFs, any signal containing energy at the relevant frequencies can be used as an excitation signal, provided it is sufficiently high against environmental noise (typically between 15 and 30 dBA) to ensure a good signal-to-noise ratio (SNR). The excitation sig-

nal’s energy must also be limited to match the dynamic range of measurement devices like loudspeakers, power amplifiers, and in-ear microphones, considering the nonlinear behavior of electro-acoustic systems. Enhancing SNR can be achieved by increasing the excitation signal length or repeating the measurement. Various methods, such as impulse excitation, stepped-sine signals, sweep signals, maximum length sequences (MLS), and others, have been proposed for measuring acoustic impulse responses, with some systematically described in the literature[18].

One of the most widespread excitation signals used in HRTF measurements is sweep signal. A sweep signal, also known as a "chirp" or "swept-sine," is a continuous signal with a frequency that changes over time and sweeps over all audible frequencies. The sweep signals are well known for identifying system nonlinearities, leading to their popularity in measuring acoustic impulse responses, such as room impulse responses, HRIRs, and electro-acoustic systems. Various types of sweep signals exist, including linear sweeps, exponential sweeps, red-colored sweeps, sweeplets, hyperbolic sweeps, and constant-SNR sweeps. Linear and exponential sweeps are particularly popular for measuring acoustic impulse responses. Sweep techniques have properties such as the ability to derive HRTFs using linear deconvolution or direct convolution with the inverse of the excitation signal. Exponential sweeps, in particular, are robust against time-variance effects and can better separate linear impulse responses from harmonic distortions.

Practical issues in the HRTF measurement process include the poor performance of loudspeakers at low frequencies and low SNRs due to limited playback levels for physiological safety. Another problem can be a measurement accuracy of person’s unique anatomical features, such as the shape of their head, pinnae, and torso. Traditional HRTF measurement methods, which involve extensive setups with multiple loudspeakers and microphones to capture sound from various angles and distances, are time-consuming and cumbersome. These setups must also maintain strict environmental controls to minimize errors caused by background noise, head movement, and other disturbances. Moreover, the process can

be particularly taxing for human subjects who must remain still for extended periods. Advances in technology have led to automated systems that aim to reduce measurement time and effort. However, these methods still face challenges in ensuring accuracy and reliability. The balance between efficiency, accuracy, and the need for controlled conditions continues to pose significant hurdles in the field of HRTF measurements. This difficulty not only makes it challenging to measure HRTFs for every individual in a mass consumer manner, but even the specifically designed and sophisticated datasets gathered in universities and special labs, as we will see later, contain serious errors and problems. This contributes to why HRTF modeling is still a somewhat unsolved problem.

1.3.2 Datasets

The field of HRTF measurement and modeling faces significant challenges due to the limitations and variability in available datasets. ML models, particularly deep learning algorithms, are data-hungry, and the current HRTF databases are often too small to meet these demands. Various research groups have built HRTF databases through acoustic measurements, with the largest database containing only 120 subjects. Unfortunately, there is no unified feature set for measuring HRTFs, leading to different datasets recording varying anthropometric features, which hinders the possibility of combining them into a single, uniform dataset. Some databases have used synthetic ear shapes to simulate HRTFs, but these are prone to domain discrepancies between the synthesized ear shapes and actual human ear shapes. These limitations contribute to the ongoing challenge of developing accurate and reliable HRTF models using ML. In the following sections, we will discuss some of the most well-known HRTF datasets in detail.

CIPIC

The CIPIC database[1], developed at UC Davis’ Center for Imaging Processing and Integrated Computing (CIPIC) Laboratory, is the most extensively utilized public-domain repository of high-spatial-resolution HRTFs. It contains head-related impulse responses for 45 subjects, measured across 1,250 spatial directions, comprising 25 azimuths and 50 elevations.

All HRTFs in CIPIC were measured with the subject positioned at the center of a hoop with a radius of 1 meter, as illustrated in Figure 1.3.2. Measurements were conducted at a resolution of 44.1 kHz and 16 bits, with HRIRs of 200 samples corresponding to 4.5 ms of audio. The sound source locations were specified by azimuth $\phi \in [-45^\circ, 230.625^\circ]$ and elevation $\theta \in [-80^\circ, 80^\circ]$ angles.

Additionally, the database provides anthropometric data for each subject along with a statistical analysis of these measurements, focusing on the temporal and spectral characteristics of the HRTFs. The anthropometric data consist of 27 measurements — 17 for the head and torso, and 10

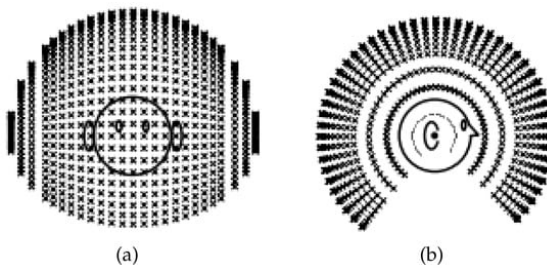


Figure 1.3.2: CIPIC Locations of data points (a) front (b) side. Adopted from [1]

Var	Measurement	μ	σ	%
x_1	head width	14.49	0.95	13
x_2	head height	21.46	1.25	13
x_3	head depth	19.96	1.29	13
x_4	pinna offset down	3.03	0.66	43
x_5	pinna offset back	0.46	0.59	54
x_6	neck width	11.68	1.11	19
x_7	neck height	6.26	1.69	54
x_8	neck depth	10.52	1.22	23
x_9	torso top width	31.50	3.19	20
x_{10}	torso top height	13.42	1.85	28
x_{11}	torso top depth	23.84	2.95	26
x_{12}	shoulder width	45.90	3.78	16
x_{13}	head offset forward	3.03	2.29	151
x_{14}	height	172.43	11.61	13
x_{15}	seated height	88.83	5.53	12
x_{16}	head circumference	57.33	2.47	9
x_{17}	shoulder circumference	109.43	10.30	19
d_1	cavum concha height	1.91	0.18	9
d_2	cymba concha height	0.68	0.12	35
d_3	cavum concha width	1.59	0.28	18
d_4	fossa height	1.51	0.33	44
d_5	pinna height	6.41	0.61	10
d_6	pinna width	2.92	0.27	18
d_7	intertragal incisure width	0.53	0.14	52
d_8	cavum concha depth	1.02	0.16	32
θ_1	pinna rotation angle	24.01	6.50	55
θ_2	pinna flare angle	28.53	6.70	47

Figure 1.3.3: Anthropometric Measurements

for the pinna. Table 1.3.3 lists all these measurements along with their statistics for all subjects.

SCUT

There are two subtypes of databases from SCUT (South China University of Technology). The first set of measurements was conducted using a KEMAR (Knowles Electronics Manikin for Acoustic Research) dummy head in an anechoic chamber. It consists of measurements from a single subject, with 10 spheres comprising $493 \times 2 \times 512$ HRIRs measured at varying distances from 0.2 to 1.0 meters. The elevation angle varies from -30° to $+90^\circ$ with a step size of 15° . The azimuth angle varies from 0° to 360° with a resolution of 5° for elevation angles between $\pm 30^\circ$. Above $+30^\circ$ elevation, the azimuthal resolution is 10° , while for $+90^\circ$ elevation, only one measurement per distance was performed. Another SCUT dataset includes measurements from 10 different human subjects and contains data $6048 \times 2 \times 512$ various positions and distances, along with anthropometric data including 9 head measurements and 15 measurements for the ear.

ARI

The ARI dataset refers to a collection of HRTFs and associated data compiled by the Acoustic Research Institute in Vienna, Austria. This dataset is widely utilized in HRTF research. Individual HRTFs were measured in a semi-anechoic chamber using a sophisticated setup of twenty-two custom-made loudspeakers. 1550 positions were measured for each listener including the full azimuthal-space (0° to 360°) and elevations from -30° to $+80^\circ$. The frontal space has a resolution of 2.5° in the horizontal plane. The speakers, driven by specialized amplifiers, played exponential frequency sweeps from 50 Hz to 20 kHz, recording via in-ear microphones placed in the subject's ears. The unique multiple exponential sweep method allowed for efficient HRTF measurement by overlapping sweeps, significantly reducing the total measurement time to about 20 minutes per subject. Directional transfer functions (DTFs) were then calculated to refine the HRTFs by filtering out common acoustic characteristics across the measured responses, ensuring high fidelity in the final HRTF data

used for auditory research. Different variations of ARI contain HRTFs for over 250 subjects. Most subjects were measured using in-the-ear microphones, while for a few others, we used behind-the-ear microphones placed in hearing aid devices. Not all subjects in ARI dataset have anthropometric data, which is only available for 60 subjects. The list of anthropometric data is align with CIPIC measurements.

RIEC

RIEC is a dataset of HRTFs measured with a circular loudspeaker array at the Research Institute of Electrical Communications, Tohoku University. The current dataset includes HRTFs for 105 subjects at 72 azimuths and 13 elevations in spherical coordinates. The vertical circular loudspeaker array consists of 35 loudspeakers arranged at intervals of 10 degrees in elevation, except for the angle directly below the subject. The circular loudspeaker array can automatically revolve around the axis of rotation with a resolution of 0.1 degree. Anthropometric data are included for 39 subjects. A three-dimensional scanner (VIVID910, Konica Minolta) was used for these measurements. All measured anthropometric dimensions are included in this dataset, in the stereolithography (STL) data format.

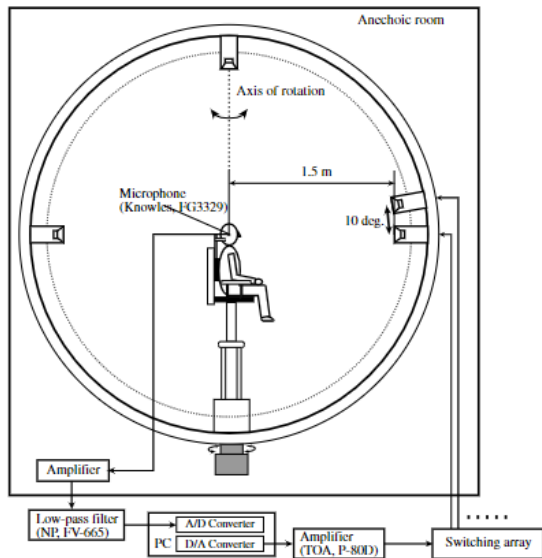
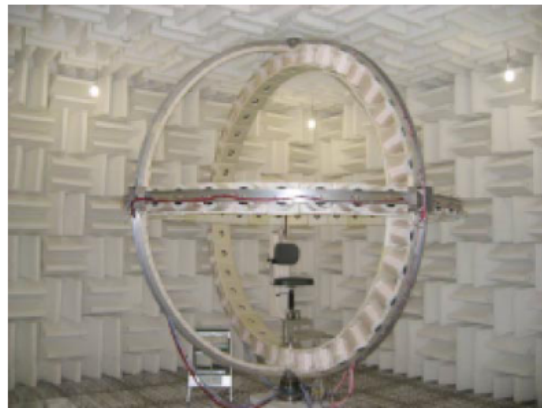


Figure 1.3.4: RIEC HRTF measurement system. Image adopted from [25]

HUTUBS

The HUTUBS [4] was developed by a collaboration involving the Audio Communication Group at the Technical University of Berlin, Huawei, and Sennheiser. The measurements were conducted in the anechoic chamber of the Technical University Berlin. The database consists of 96 subjects, including 93 human subjects and the FABIAN head-and-torso simulator. For evaluation purposes, FABIAN and one human subject were measured twice. The database includes 10 female and 83 male participants. The dataset includes both acoustically measured and numerically simulated HRTFs for each

subject. Measured HRTFs were acquired in an anechoic chamber using a setup involving 37 loudspeakers positioned in a cylindrical array around the subject. The subjects were rotated on a turntable to capture HRTFs from different directions, resulting in measurements at 440 spatial positions. Simulated HRTFs were generated using numerical simulations based on high-resolution 3D head meshes of the subjects. The simulations were calculated for frequencies between 100 Hz and 22 kHz.

The dataset includes detailed anthropometric data for each subject, encompassing 25 different features measured from the head, torso, and ears. Measurements for the head and torso were taken directly using tape measures, while pinna features were either estimated from photographs or derived from 3D surface meshes of the ears. Specific anthropometric features in the dataset include dimensions such as head width, head height, head depth, neck dimensions, and detailed measurements of the pinnae, like cavum conchae height, cymba concha

	feature	mean (SD)	var.
x_1	head width ^a	15.24 (0.72)	9.44
x_2	head height ^a	20.96 (1.04)	9.92
x_3	head depth ^a	20.24 (0.83)	8.20
x_4	pinna offset down ^a	0.27 (0.19)	140.74
x_5	pinna offset back ^a	0.39 (0.32)	164.10
x_6	neck width ^a	12.13 (0.91)	15.00
x_7	neck height ^a	7.02 (1.30)	37.03
x_8	neck depth ^a	11.96 (1.33)	22.24
x_9	torso top width ^m	33.64 (3.52)	20.92
x_{12}	shoulder width ^m	54.80 (4.46)	16.27
x_{14}	height ^m	181.03 (8.28)	9.14
x_{16}	head circumference ^m	56.80 (1.97)	6.93
x_{17}	shoulder circumference ^m	115.11 (8.67)	15.06
d_1	cavum conchae height ^a	1.81 (0.16)	17.67
d_2	cymba concha height ^a	1.01 (0.11)	21.82
d_3	cavum concha width ^a	1.75 (0.19)	21.71
d_4	fossa height ^a	2.10 (0.24)	22.85
d_5	pinna height ^a	6.14 (0.40)	13.02
d_6	pinna width ^a	2.97 (0.26)	17.50
d_7	intertragal incisure width ^m	0.63 (0.14)	46.39
d_8	cavum concha depth (down) ^m	1.15 (0.13)	23.08
d_9	cavum concha depth (back) ^m	1.19 (0.13)	22.07
d_{10}	crus helix slant height ^m	0.31 (0.06)	39.12
Θ_1	pinna rotation angle ^s	10.50 (4.60)	87.61
Θ_2	pinna flare angle ^s	25.35 (7.24)	57.12

Figure 1.3.5: Anthropometric features from HUTUBS database. Table taken from [4]

Name	Subjects	Directions	Spatial sampling	Room	Anthropometry	3D meshes	Simulated HRTFs	Spatially continuous
CIPIC [4]	45	1250	$\Delta\phi = 5^\circ; \Delta\theta = 5.6^\circ$ $-45^\circ \leq \theta \leq 80^\circ; r = 1 \text{ m}$	room with absorbers	43 subjects	–	–	–
LISTEN [5]	51	187	$\Delta\phi = 15^\circ; \Delta\theta = 15^\circ$ $-45^\circ \leq \theta \leq 90^\circ; r = 1.95 \text{ m}$	anechoic	yes	–	–	–
FIU [6]	15	72	$\Delta\phi = 30^\circ; \Delta\theta = 18^\circ$ $-36^\circ \leq \theta \leq 54^\circ$	room with absorbers	yes	pinnae only	–	–
ARI [7]	150	1550	$\Delta\phi = 2.5^\circ, 5^\circ; \Delta\theta = 5^\circ$ $-30^\circ \leq \theta \leq 80^\circ; r = 1.2 \text{ m}$	hemi-anechoic	60 subjects	–	–	–
RIEC [8]	105	865	$\Delta\phi = 5^\circ; \Delta\theta = 10^\circ$ $-30^\circ \leq \theta \leq 90^\circ; r = 1.5 \text{ m}$	anechoic	39 subjects	head and shoulder	–	–
SYMARE [9]	10	393	$\Delta\phi \approx 10^\circ \text{ GCD}^1; \Delta\theta = 10^\circ$ $-45^\circ \leq \theta \leq 90^\circ; r = 1 \text{ m}$	anechoic	yes	head and shoulder	up to 16 kHz	–
ITA [10]	48	2304	$\Delta\phi = 5^\circ; \Delta\theta = 5^\circ$ $-66^\circ \leq \theta \leq 90^\circ; r = 1.2 \text{ m}$	hemi-anechoic	yes	pinnae only	–	–
HUTUBS ² (this work)	96	440	$\Delta\phi \approx 10^\circ \text{ GCD}^1; \Delta\theta = 10^\circ$ $-90^\circ \leq \theta \leq 90^\circ; r = 1.47 \text{ m}$	anechoic	yes	head with-out shoulder	up to 22 kHz	yes

Figure 1.3.6: Public HRTF databases that include anthropometry in chronological order of publication. Table taken from [4].

height, pinna height, pinna width, and intertragal incisure width. High-resolution 3D head meshes were acquired using a Kinect 3D scanner for the head and a high-resolution Artec Space Spider scanner for the ears. These meshes were used to derive precise anthropometric measurements and ensure accurate simulation of HRTFs. The features were extracted automatically or semi-automatically where possible, reducing biases and uncertainties associated with manual measurements. This comprehensive set of anthropometric data, alongside the HRTF measurements and simulations, provides a valuable resource for detailed analysis and research.

The HUTUBS dataset is considered one of the more comprehensive and high-quality datasets available for HRTF research in the open access today.

Summary

The summary of HRTF datasets is given in the Figure 1.3.6. The table on the figure provides an overview of various HRTF datasets, detailing their key characteristics. The datasets vary in the number of subjects, ranging from 10 (SYMARE) to 150 (ARI), and the number of spatial directions sampled, with ITA having the highest at 2304 and FIU the lowest

at 72. The spatial sampling methods and resolutions differ significantly, with some datasets like ARI and ITA offering high spatial resolutions. Most datasets were measured in anechoic or hemi-anechoic rooms to ensure accuracy. However, some of the most popular datasets, like CIPIC, were collected in rooms with absorbers, which would introduce many inaccuracies in the dataset. Anthropometric data is included in most datasets, though the extent varies; for example, CIPIC includes data from 43 subjects, while HUTUBS provides comprehensive anthropometric and 3D mesh data for all subjects. Additionally, some datasets like HUTUBS also offer simulated HRTFs. These variations highlight the problems of dataset diversity and the differing approaches and data quality across different HRTF datasets. This is why combining these datasets to obtain one comprehensive dataset has not still been achieved.

1.3.3 Evaluation of HRTFs

Evaluating HRTFs in ML models involves several key metrics to ensure accurate and perceptually relevant results. These metrics can be categorized into objective and subjective metrics. Objective metrics typically include spectral distortion, which measures the deviation between the frequency responses of the original and estimated HRTFs. Other variation cans be mean squared error (MSE) and root mean squared error (RMSE), which compute the average squared difference and the average magnitude of the error, respectively. Subjective metrics are based on human perception and typically involve listening tests, where subjects evaluate the spatial accuracy and quality of audio rendered using the estimated HRTFs. Combining these objective and subjective metrics provides a comprehensive evaluation of HRTFs in ML models, ensuring both technical accuracy and perceptual relevance.

One of the most commonly used metrics for objective evaluation is the log spectral distortion (LSD), which compares the generated HRTFs to the true HRTFs by measuring the spectral differences. The LSD is calculated as:

$$LSD(y, \hat{y}) = \sqrt{\frac{1}{N} \sum_{i=0}^{N-1} \left(20 \log_{10} \frac{y[i]}{\hat{y}[i]} \right)^2}$$

where $y[i]$ and $\hat{y}[i]$ are the true and predicted HRTF magnitudes at the i -th frequency bin, respectively, and N is the number of frequency bins. Typically, an LSD value around 4 dB is considered imperceptible in listening tests, indicating good model performance. Additionally, some similar metric are used in some works as mean squared error (MSE), wich is similar to this LSD and might only have scaling difference.

Subjective evaluation experiments play a crucial role in assessing the performance of HRTFs. Some works conduct subjective evaluation experiments. Usually, localization accuracy is assessed by asking subjects to point out the perceived direction of sounds, and perceptual evaluation of audio quality (PEAQ) is used as a standardized method to evaluate the perceived audio quality. Ofter, subjective localization experiments for evaluating HRTFs involve comparing the perceived directions of sounds filtered by measured and estimated HRTFs. These experiments often use bursts of white noise presented at various azimuths, with subjects indicating the perceived direction of each sound. The setup includes reference sounds to help orient the subject. Variations of these experiments include using different types of stimuli, azimuth intervals, and methods for presenting and collecting feedback, such as calling out azimuths or using visual aids. Some studies may also involve more dynamic or complex auditory scenes to evaluate the robustness of the HRTFs under different conditions. Additionally, the experiments may assess other perceptual qualities like externalization, distance perception, and timbral fidelity. These comprehensive evaluations help in understanding how well the HRTFs can replicate realistic spatial audio perception and identify areas for improvement in HRTF estimation techniques.

1.4 Overview of Existing Literature/Work

There has been a tremendous amount of work dedicated to HRTFs modeling with ML in the last two decades, with many advancements occurring especially in the last decade as ML technology progressed and proposed solutions to numerous problems. The promise to solve nonlinear and complex problems has greatly influenced the field of HRTFs, leading to the use of ML models to address its standing issues, as discussed earlier (see: 1.2.3). There are several primary directions related to solving problems in acquiring HRTFs in different scenarios considering problems HRTF research faces, which we will explore in subsequent sections. Broadly, these works can be categorized into three distinct groups: (1) Selecting Matching HRTFs, (2) Interpolation/Extrapolation - Up-Sampling of HRTF Sets, and (3) Synthesis/Generation of HRTFs from Anthropometric Data.

1.4.1 Selecting Matching HRTFs

The main idea behind selecting matching HRTFs is to avoid making costly and complex measurements for every individual by using HRTFs from a matching or anthropometrically similar person, which would work as well or nearly as well as individually measured HRTFs. This approach addresses the challenge of selecting the most appropriate HRTFs from existing datasets to create optimal virtual acoustic spaces in binaural audio systems without extensive measurements and listening tests. It involves using neural networks or regression methods to identify the most suitable HRTF set for an individual from a large database. These systems often use anthropometric data as input features to predict a matching score for the existing HRTF data points, allowing for the selection of the closest match.

In the past, researchers tackled the problem by employing different statistical methods. For instance, the paper "A hybrid algorithm for selecting head-related transfer function based on similarity of anthropometric structures" [30] by Xiangyang Zeng, Shu-guang Wang, and Lipan Gao (2010) introduces a hybrid algorithm that integrates Principal Component

Analysis (PCA), Multiple Linear Regression (MLR), and Database Matching (DM) for the individualization of HRTFs. This research uses CIPIC as matching dataset and leverages the similarity and relativity of anthropometric structures to quickly and effectively personalize HRTFs. The hybrid algorithm’s efficacy was tested through subjective listening experiments comparing the localization accuracy and front-back confusion issues using both best and worst match-selected HRTFs for binaural auralization. The study concludes that their algorithm significantly improves localization accuracy and reduces front-back confusion, showcasing its potential for enhancing spatial audio experiences based on the listeners’ unique anthropometric features.

Another study from 2012, "Perceptually based head-related transfer function database optimization" [12] focused on a similar approach with human perceptual optimisation but with additional step of reducing HRTF database size. The study employed subjective ratings from listeners to create a perceptually optimized subset of HRTFs. The optimized subset was validated through localization tests, showing reduced localization errors when subjects used their best-matched HRTF compared to their worst match. This work illustrated the potential of using listener feedback to refine HRTF selection and database management for improved spatial audio rendering.

One of the most well known studies in the direction on selecting matching HRTFs is the paper titled "Head-Related Transfer Function Selection Using Neural Networks" by Shu-Nung Yao, Tim Collins, and Chaoyun Liang (2017) [29]. The study introduces a neural network that estimates the suitability of HRTF datasets using anthropometric measurements as input vectors. These measurements, obtained from digital photography of listeners’ heads and pinnas, form the feature input vectors for the neural network. Researchers used CIPIC dataset with its anthropometric parameters for matching. A graphical user interface (GUI) was developed for participants to evaluate the fitness of each HRTF dataset by listening to music transformed with different HRTFs. The scores from these evaluations were used as target outputs to train the neural networks, aiming to learn a mapping between anthropo-

metric parameters and listeners’ perception scores. Experimental validation with 30 subjects showed that this system offers a more reliable HRTF selection method compared to previous statistical methods based on the sum of the squared error between pinna parameters. The performance was evaluated by calculating how many people would be offered their ‘favourite’ HRTF datasets by using the method.

Robert Pelzer et al proposed a machine learning model method aiming to recommend HRTFs that provide a clear improvement over generic or randomly selected HRTFs in their 2020 study ”Head-related transfer function recommendation based on perceptual similarities and anthropometric features” [23]. The approach demonstrated the effectiveness of matching HRTFs based on perceptual and physical similarity, highlighting the relevance of anthropometric features in the selection process.

A recent paper ”Towards Fast And Convenient End-To-End HRTF Personalization” [31] by Bowen Zhi, D. Zotkin, and R. Duraiswami, published in 2022, introduces a system that employs a more sophisticated UNet convolutional neural network model to predict pinna landmarks from photos of ears. The pinna landmarks is used to extract anthropometric features used for HRTF personalization. These anthropometric features then are matched to the HRTF from existing dataset with the nearest-neighbor matching. Table 1.4.1 is a main results table from the paper showing how the method works for CIPIC dataset.

Even though the HRTF matching direction on the research studies showcased significant results and some of their methodologies may have been adopted and are likely still employed in real-life technology solutions—especially in industries focusing on

	RMSE (dB)	LSD (dB)
Ours	-25.957 ± 6.513	5.315 ± 3.154
Average	-25.226 ± 6.360	7.208 ± 2.238

Table 1.4.1: HRIR approximation error on the CIPIC database, comparing the HRIRs computed using our method with an average HRIR. HRIR comparison is done on a per-ear basis, across all available azimuths and elevations.

virtual reality (VR), augmented reality (AR), gaming, and spatial audio due to their convenience—the methods are far from being precise and robust for everyone. Effective application typically requires that the anthropometric data of an individual closely matches with a given

data point within the dataset. Moreover, most of this research relies on subjective evaluations as metrics, due to the absence of a precise mathematical way to measure the effectiveness of HRTF customization. This subjective nature means that the outcomes are not directly quantifiable like such a metrics as LSD and other precise mathematical ones.

Despite these limitations, HRTF customization methods offer a fast and cost-effective solution, delivering relatively good results with ease of use. While specific company details may not always be publicly disclosed due to proprietary technologies and competitive advantages, there's indirect evidence of industry adoption of similar principles to enhance spatial audio experiences. Arguably, companies like Dolby, Sony, Embody, and Waves Audio, among others, have been (and most probably still are) employing matching personalization algorithms in their audio, gaming, and multimedia applications.

1.4.2 Interpolation/Extrapolation - Up-Sampling of HRTF Sets

The Interpolation/Extrapolation, or in other words, the up-sampling of HRTF Sets approach tackles a different aspect of HRTFs—specifically, the challenge of approximating HRTFs at missing positions or reconstructing a full set from a limited subset of HRTFs. The rationale in this approach is that no measurement can capture HRTFs for all possible continuous positions, and we always need to interpolate between them, whatever the precision of the data we have. Additionally, in some scenarios we can have only partial data from the individual HRTFs as these sparse measurements can be obtained more easily, quickly, and at a lower cost compared to full measurement data. These methods often employ fully connected or convolutional neural networks. More recent studies demonstrate the extensive use of Variational Autoencoders (VAE). These techniques utilize sparse HRTF measurements as prediction inputs, sometimes incorporating anthropometric measurement data too and give HRTFs for missing positions for a given individual.

One of the earliest works in the direction of up-sampling HRTF data is described in the paper "Individualized HRTFs from few measurements: a statistical learning approach" [17] from 2005. The paper discusses a method for creating virtual auditory spaces (VAS) with high fidelity using individualized HRTFs based on minimal user measurements. The authors propose statistical modeling techniques that allows for individual HRTF synthesis with only a few measurements for each user, potentially reducing the costs associated with 3-D audio systems. The proposed approach involved integrating a collection of uniformly distributed HRTFs with a Self-Organizing Maps (SOM)[14] and neural network-based model. Using CIPIC dataset, this work demonstrated the potential to accurately synthesize a user's HRTFs across all orientations by relying on just 50 measurements from the individual, achieving an average modeling error of less than 3 dB.

The paper from 2014, "HRTF magnitude synthesis via sparse representation of anthropometric features" [3] introduces a method for synthesizing the magnitudes of HRTFs by leveraging a sparse representation of an individual's anthropometric features and a small

subset of given HRTF. Sparsity of anthropometric features in the context is achieved by a method of optimizing an objective function to balance signal reconstruction accuracy, thereby enhancing robustness against signal corruptions such as noise, missing data, and outliers, as described in [10]. In this study, researchers collected their own dataset of HRTFs of 96 anthropometric features of 36 subjects and employed Absolute Shrinkage and Selection Operator (LASSO) [15] method, a kind of a regression analysis for finding a sparse Representation of anthropometric data for consequent HRTF Synthesis. The method was validated by LSD comparison between the synthesized and the actual measured HRTFs for the subject. The proposed method demonstrated lower distortions compared to other evaluated techniques (Ridge Regression[8] and HATS - a manikin that is designed based on average anthropometric features) and achieves results that are nearly as accurate as the best possible HRTF classifier (i.e., the nearest HRTF in the training set).

A 2018 paper “Subspace-based hrtf synthesis from sparse data: A joint pca and ml-based approach,” [2] by S. G. Bharitkar, T. Mauer, T. Wells, and D. Berfanger, presents a method for synthesizing HRTFs from sparse data, specifically for the IRCAM-Listen HRTF database. It employs auditory localization blur to augment sparse data and uses PCA to reduce the dataset to six principal components. These components are then used to train an Artificial Neural Network (ANN), which outputs the principal components corresponding to arbitrary HRTF directions. Unfortunately, this paper is not in open access, and further details are limited as the review is based on the abstract.

A 2019 paper titled ”Head Related Impulse Response Interpolation and Extrapolation Using Deep Belief Networks” [13] by G. Kestler, S. Yadegari, and D. Nahamoo explores a machine learning technique using Deep Belief Networks (DBNs) to handle both interpolation and extrapolation of HRTF databases. The network architecture from this work is shown in Figure 1.4.1, depicting only the left side of the network with identical architecture for the right side. To accommodate the small amount of data, reserchers proposed a cascaded architecture of separately trained fully connected sub-networks each utilising the particular

component of physics behind the HRTFs. The researchers decoupled the stereo pair of HRTFs for each ear and removed inter-aural time differences and distance attenuation. This refinement allows the networks to focus solely on interpolating and extrapolating the spectral filtering of audio for both ears. The methodology was tested on the CIPIC and SCUT databases, yielding an average log spectral distortion of less than 3 dB for all tested points around the head, indicating a high level of accuracy in the spectral domain for these auditory models.

This work has been continued and extended for the direct prediction of HRTFs from anthropometric data, which is presented in the experiment section of this writing (see Section 1.5).

More recent paper "Deep HRTF Encoding Interpolation: Exploring Spatial Correlations Using Convolutional Neural Networks" [33] by D. Zurale, S. Yadegari, and S. Dubnov, delves into the application of deep learning, specifically Convolutional Neural Network based auto-encoding strategy for HRTF encoding and then we use the learned auto-encoder to provide an alternate solution for the interpolation of HRTFs from a sparse dataset of HRTFs. The works shows that these method is capable of achieving results that are comparable to other non deep learning based approaches, in spite of using only a few tens of data points.

Similar problem was tackled by another paper form 2023 "Modeling individual head-related transfer functions from sparse measurements using a convolutional neural network" [11] by Z. Jiang, J. Sang, C. Zheng, A. Li, and X. Li. This research used HRTFs' sparse

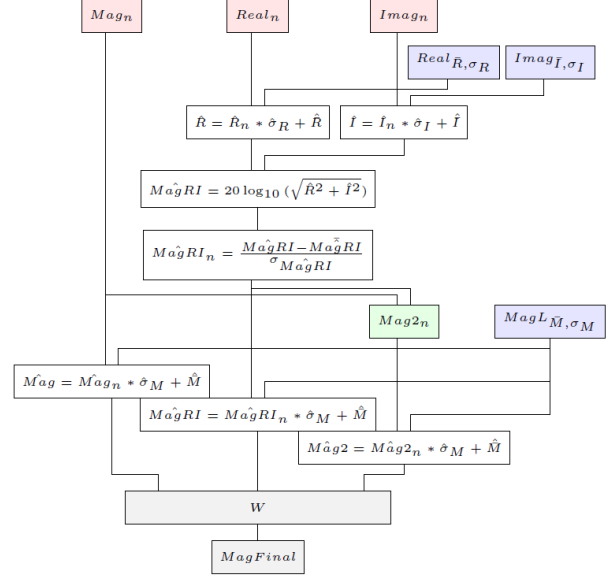


Figure 1.4.1: Red and Green networks predict zero mean, unit variance normalized functions. Networks in purple predict the mean and standard deviation of the raw value to be used for un-normalization of the Red and Green predictions later. The white blocks indicate mathematical operations that were not trained in the network. The mixing matrix W and the $MagFinal$ network produce the final magnitude output.

measurements represented as two-dimensional images and a convolutional neural network (CNN). The CNN was trained with sparse HRTF images as inputs and corresponding high-resolution images from a pre-existing HRTF database as outputs. The study used CIPIC dataset and demonstrated that using only 23 directions from 1250 resulted in a spectral distortion (SD) of about 4.4 dB, which decreased to approximately 3.8 dB when using 105 directions.

The paper titled "Spatial Upsampling of Head-Related Transfer Functions Using a Physics-Informed Neural Network" [20] explores using a physics-informed neural network (PINN) to upsample sparse Head-Related Transfer Function (HRTF) data. The approach leverages the Helmholtz equation to guide the network, ensuring the upsampling adheres to physical acoustic principles, which helps avoid overfitting and enhances model generalization. This method aims to produce more accurate and physically plausible HRTF interpolations, enhancing virtual acoustic applications' realism.

Although these techniques often yield promising results and strong performance metrics, their reliance on existing HRTF measurements poses a significant drawback. This dependence renders them more of an intermediate solution rather than a direct path to ultimate goal of obtaining personalized HRTFs from easily accessible data, limiting their applicability for fully personalized auditory experiences.

1.4.3 Synthesis/Generation of HRTFs form Anthropometric Data

The Synthesis/Generation of HRTFs from Anthropometric Data using ML and deep learning methods stands out for its universality, aiming to directly create HRTFs for any individual and position, considering personalization needs. The process often involves learning and using meaningful representations of HRTFs, with an objective of dimensionality reduction to aid more effective prediction of individualized HRTFs by downstream generative models. Extraction of representations often include PCA, as well unsupervised learning of HRTFs features by such models as Autoencoders, based on a variety of structures such as the Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), or Vector Quantized Variational Autoencoder (VQ-VAE). The input data for these methods is also varied, including anthropometric measurements, images/videos of ears and torso, and other variations of personal data. Moreover, these techniques frequently involve extracting predictors from user data. This includes compressing and choosing the most important ones from anthropometric data through PCA analysis, processing of ear and torso images through networks (Convolutional autoencoders for example to find meaningful latent representations), and other methodologies. While this method could potentially offer a universal solution for audio spatialization, to date, none of the existing research has demonstrated outstanding results. This limitation can largely be attributed to the small sizes of existing datasets and often occurred errors and non-standardised methods of data acquisition in them. Deep learning models used in this type of methods would undoubtedly benefit from access to larger datasets that include more subjects. Nevertheless, the effort expended and knowledge acquired in this direction is expected to lead to more accurate models for HRTF individualization in the future.

One of the earliest attempts to enhance the realism of virtual auditory spaces (VAS) by personalizing head-related transfer functions (HRTFs) using artificial neural networks (ANNs) is documented in the paper titled “HRTF Personalization Based on Artificial Neural Network in Individual Virtual Auditory Space,” [9] published in *Applied Acoustics* in 2008.

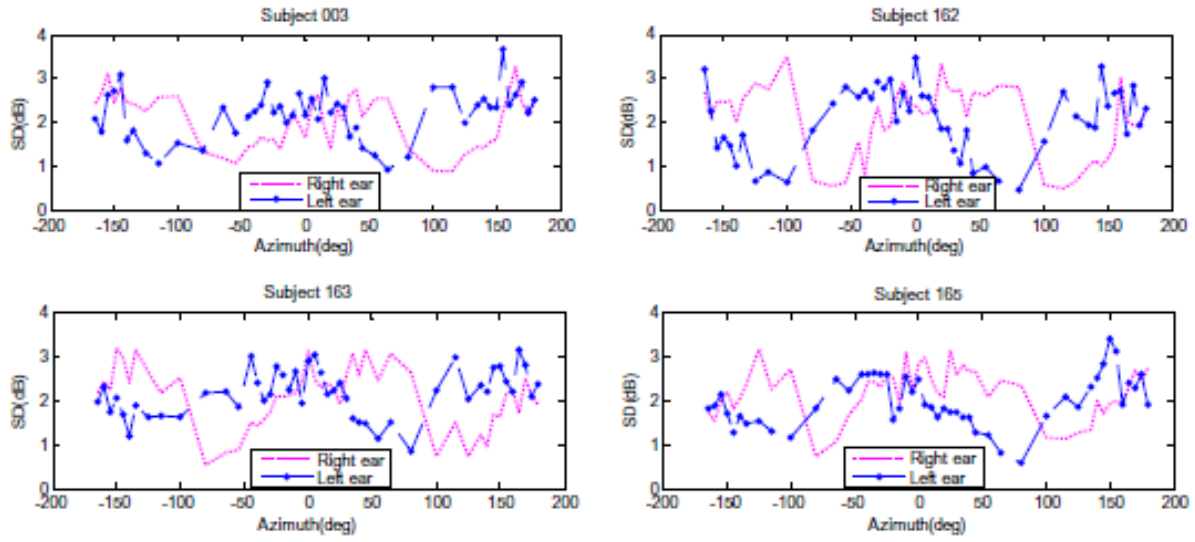


Fig. 7. Left and right ears SD scores by azimuth in the horizontal plane.

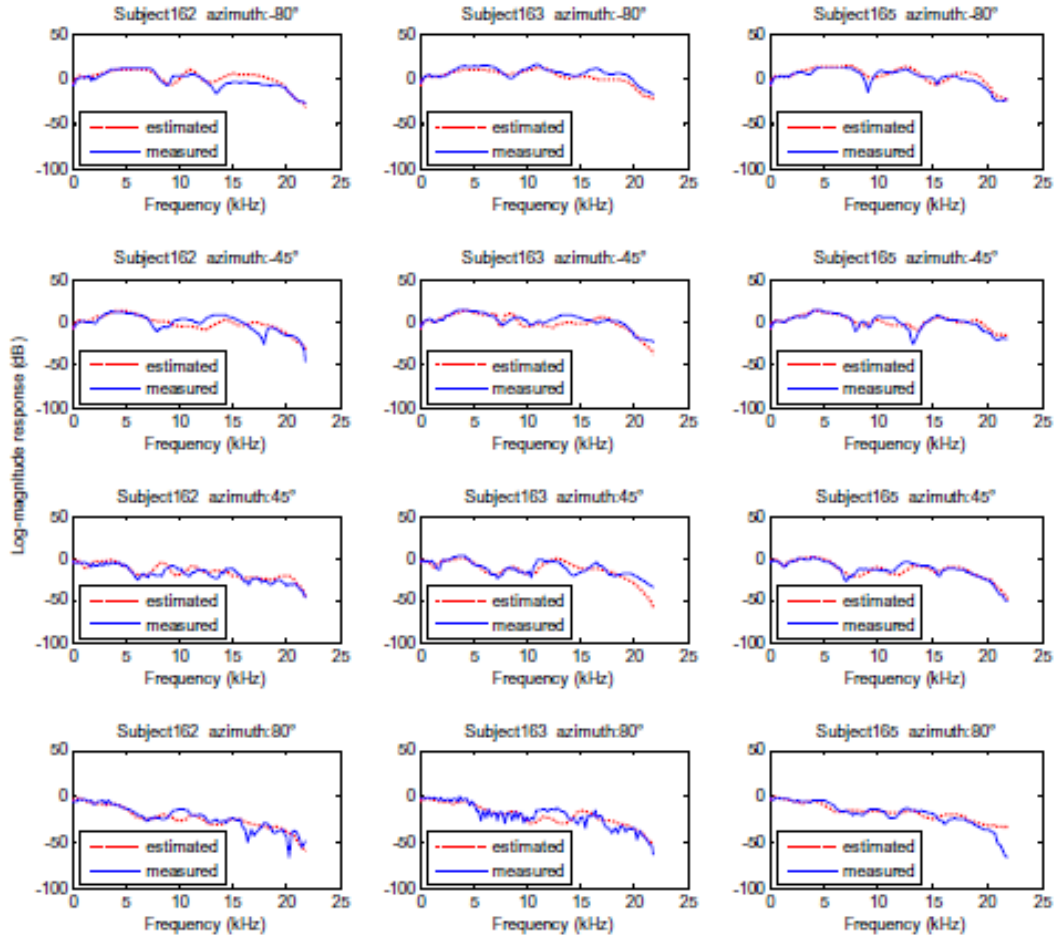


Fig. 8. Log-magnitude response of the estimated and measured HRTF.

Figure 1.4.2: Results page from [9]

This study proposes a customization method employing a fully connected neural network, where the inputs are anthropometric parameters identified through PCA for their significant correlation with HRTF characteristics. Utilizing data from the CIPIC database, the study selects eight key anthropometric variables critical for HRTF formation: head width (x_1), head depth (x_3), shoulder width (x_{12}), torso top width (d_1), cavum concha width (d_3), fossa height (d_4), pinna height (d_5), and pinna width (d_6). The performance of this method was evaluated using both objective metrics and subjective sound localization tests. Objective evaluations aimed to compare the effectiveness of the personalized predicted HRTFs with those obtained from standard databases. The results indicated spectral distortion below 3 dB at zero elevation plane; however, the paper only reported results for zero elevation plane and it does not explicitly specifies whether these findings were derived from the test subjects or not. Figure 1.4.2 show results page from the paper. As we see in the figures the estimation lines lack significant details compared to real measured HRTFs. Furthermore, the study does subjective evaluation tests, demonstrating improvements in localization accuracy with personalized HRTFs. Specifically, the correct estimation of sound direction increased from 55% to 75%, and front-back confusion decreased from approximately 12-13% to 9.7% for study subjects with proposed model compared to generic HRTF model.

Later in a paper "On Data-Driven Approaches to Head-Related-Transfer Function Personalization" [7] suggestion was reported that a statistical approach such as PCA fails to encode features that are essential to the individuality of the HRTFs and the models trained on full HRTFs improve performance.

In the 2017 paper titled "Deep Neural Network Based HRTF Personalization Using Anthropometric Measurements" [6] by C. J. Chun, J. M. Moon, G. W. Lee, N. K. Kim, and H. K. Kim, pulished in authors propose a method for personalizing HRTFs using deep neural networks (DNNs) and anthropometric measurements. They utilized the CIPIC HRTF database to develop a DNN model that predicts an individual's HRIR (head-related impulse response) samples for the left ear based on measurements of their head, torso, and pinna. The

Azimuth	Average HRTF	DNN(37) HRTF	DNN(17) HRTF	CNN-DNN HRTF
-135°	11.29	4.12	6.62	5.70
-80°	4.12	3.12	4.05	3.11
-45°	7.41	3.82	5.30	4.61
Avg.	7.61	3.69	5.32	4.47

Table 1.4.2: comparison of models in [16]

proposed personalization approach was evaluated using root-mean-square error (RMSE) and log-spectral distortion (LSD) metrics. The results indicated that the personalized HRIRs estimated by the DNN model were more accurate than the generic HRIRs averaged across all subjects in the database, showing lower values in both RMSE and LSD. This research demonstrates the potential for improving spatial audio rendering by tailoring HRTFs to individual listeners’ physical characteristics. Unfortunately, the paper is not in open access, so we weren’t able to access all the details of implementation and results. Nevertheless, we mention this paper because it has been used as a baseline for few following works.

In a subsequent paper ”Personalized HRTF Modeling Based on Deep Neural Network Using Anthropometric Measurements and Images of the Ear”[16] same authors proposes a novel estimation method for personalized HRTFs utilizing deep neural networks (DNNs). This method uses images of

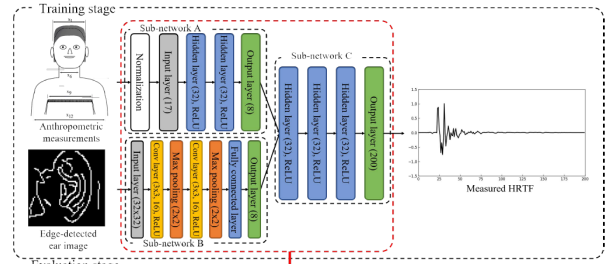


Figure 1.4.3: Illustration of proposed model in [16]

the ear in addition to anthropometric measurements. Authorts used fully connected and convolutional layers to deal with mix data of images and measurments. Figure 1.4.3 shows an architecture of the proposed model. The evaluation of the method included calculating RMSE and LSD between the reference and estimated HRTFs, as well as subjective evaluations. Researchers compared four different models: the Average HRTF without personalization, a model trained solely on anthropometric data ”DNN(37) HRTF” (the same model as in [6]), a model trained on anthropometric data excluding ear measurements ”DNN(17) HRTF”, and the proposed model utilizing both images and anthropometric data ”CNN-

DNN HRTF”. As they have shown in the paper and we can see in Table 1.4.2, the proposed method demonstrated an RMSE of -18.40 dB and an LSD of 4.47 dB. These figures are 0.02 dB lower for RMSE and 0.85 dB higher for LSD than those for the DNN-based method that uses anthropometric data without pinna measurements, respectively. However, in the paper the results were given only for three positions: -135° , -80° , and -45° degrees. Moreover, the paper, for some reason, provides averages for all subjects, not discriminating between training to test subjects. The same issue arises with the RMSE and LSD graphs presented in the paper; they merely provide numbers for all CIPIC subjects. In terms of subjective evaluation, the paper reports that with proposed method test subjects were able to localize sound sources with an increased accuracy of around 11% and 6% compared to the Average HRTF method and the DNN-based method, respectively. Additionally, the proposed method achieved reductions in the front/back confusion rate by 12.5% and 2.5% when compared to the Average HRTF method and the DNN-based method, respectively.

In the paper ”Autoencoding HRTFs for DNN-Based HRTF Personalization Using Anthropometric Features” [5] by Tzu-Yu Chen, Tzu-Hsuan Kuo, and T.-S. Chi

Models	Baseline	AutoEn + DNN	Joint Training	Dataset Comb.
mean LSD	3.705	3.429	3.252	3.246

Table 1.4.3: Mean LSD of compared approaches over all azimuth angles with elevation 0

(2019), the authors propose a two-level deep neural network Variations Auto-encoder (VAE) + fully connected Neural network (DNN) based approach to synthesize personalized HRTFs using the user’s anthropometric features. The study uses autoencoder for dimensionality reduction of HRTF data and finding a critical set of features that represent the raw HRTFs. The use of VAE reduced feature set of HRTFs that was subsequently learned by a smaller DNN. This approach, as authors claim, helps prevent overfitting. The research utilized the CIPIC dataset; however, in additional experiments, the autoencoder was trained on a extended dataset including HRTFs from other data too, learning a more general representation of HRTFs. The study focused on magnitude HRTFs between 200 Hz and 15 kHz and normalized to values between 0 and 1. VAEs were trained for every elevation, and separate

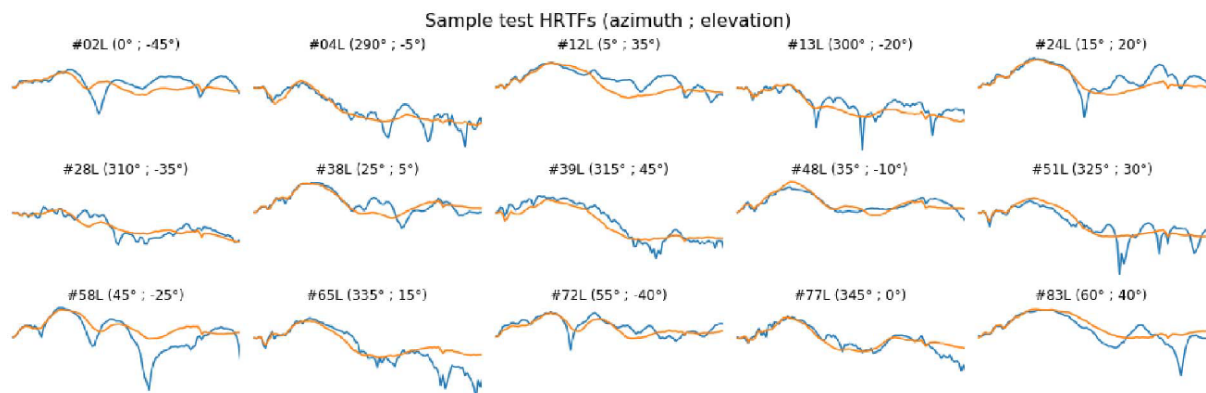


Figure 1.4.4: Results page from [21]

DNNs were trained for every azimuth angle. VAEs and DNNs were trained separately but also experiment was conducted with joint fine-tuning. The effectiveness of this method was compared against a baseline [6] DNN model without an autoencoder. Illustrated in the table 1.4.3, the results of the study were presented only for elevation 0 points using LSD metric.

The paper titled "HRTF Individualization using Deep Learning" [21] by Riccardo Miccini and Simone Spagnol, published in 2020, provides a survey-like exploration into the individualization of Head-Related Transfer Functions (HRTFs) using deep learning techniques. The paper discusses three primary research directions. Firstly, the extraction of predictors from user data, focusing on identifying pertinent information that can predict individual HRTFs. Secondly, it discusses the unsupervised learning of HRTFs, showcasing the use of autoencoder networks to learn HRTFs without the need for supervised training data. Lastly, the synthesis of HRTFs from anthropometric data is explored, employing deep multilayer perceptrons and PCA to generate HRTFs based on the physical characteristics of users. Different from previous papers discussed here, this paper reports LSD score for all positioned and clearly distinguished train and test subjects. For HRTF generation from anthropometric data the best model from this study shows LSD score of 4.1 for train and 4.7 for test subsets. Some of the HRTF reconstructions are presented in the figure 1.4.4. As we may see reconstructions are not very good, they lack details and are approximating overall average contour shape of the HRTF.

A More exotic approach is taken the work "Fully perceptual-based 3D spatial sound individualization with an adaptive variational autoencoder" [28] by Kazuhiko Yamamoto and T. Igarashi (2017). This study investigates the application of a Variational Autoencoder (VAE) equipped with a Convolutional Neural Network (CNN) for the individualization of 3D spatial sound. Initially, the Autoencoder is trained using the publicly available CIPIC HRTF dataset. Subsequently, a Human-in-the-loop setups have been proposed for the perceptual fine-tuning process. The process involves human feedback to fine-tune the latent variable serving as input to the model's generative component. The researchers also employed a novel approach to data handling, wherein the input data is transformed into HRTF patches. Each patch is a 4-dimensional tensor with the dimensions $(5 \times 5 \times 128 \times 4)$, where the first two dimensions represent the HRTF under investigation and its neighbors along the elevation and azimuth directions, encompassing a total of 25 HRTFs per patch. The latter dimensions detail each HRTF's content within the patch: the final dimension, also referred to as the channel, encodes the frequency power spectrum or time-domain signal for either the left or right ear, with 128 representing their length. This form of data representation furnishes a significant amount of contextual information, which can be effectively learned by the 3D convolutional layers. Study was evaluated for 20 participant and against the best matching CIPIC HRTFs and method proved to be superior in all 20 cases with substantial improvement.

A more recent paper from 2021, "On the Predictability of HRTFs from Ear Shapes Using Deep Networks," [32] authored by Yaxuan Zhou, Hao Jiang, and Vamsi Krishna Ithapu from Facebook Reality Labs, focuses on the challenge of HRTF synthesis using 3D ear shape images as inputs. It explores the bounds of HRTF predictability using deep neural networks. The paper proposes and evaluates two models, CNN-Reg and UNet-Reg, a convolutional neural network and a well known UNet [24] architecture that proved to be successful in the variety of field of ML. The study utilized HRTFs from 645 human subjects collected by Artec 3D

scanners. Researchers approached data scarcity problem by generating data themselves. First, the 3D meshes of the head and upper torso for subjects were obtained. Then, The left-ear and right-ear HRIRs on a 1-meter sphere were simulated using the acquired 3D mesh with the finite-difference time-domain (FDTD) method. As the authors claim, since the low frequency content in HRTFs does not contain notches or peaks and simulations at high frequencies

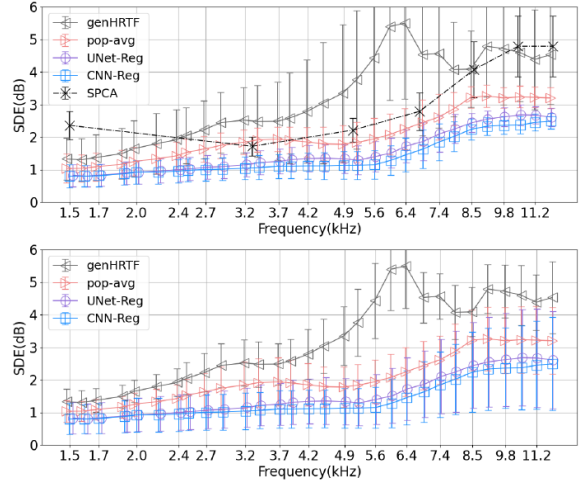


Figure 1.4.5: Mean and s.d of SDE vs. frequencies. (Top) s.d across directions; (Bottom) s.d across subjects in [32]

are noisy, they focused on the 30 frequency bins between 1 and 12 kHz. HRTFs were simulated with a step of 10° . The azimuth has 36 positions in the range of $(0^\circ, 360^\circ)$ and the evaluation was restricted to 10 positions across elevations $(-30^\circ, 60^\circ)$, resulting in a total of 360 positions. 3D ear images were cropped with center around the ear canals and used as 3D tensors for networks. Additionally, to increase dataset authors mirrored all right ear-HRTF pairs to form new left ear-HRTF pairs, which increases the dataset size to a total of 1290 ear-HRTFs pairs. The method used 30 separate networks, one per frequency bin, with output size of 360 giving numbers for each direction. In the figure 1.4.5, the results of CNN-Reg and UNet-Reg are compared to population average (denoted by popavg) of data and a simulated HRTF on KEMAR dummy head (denoted by genHRTF). Both plots demonstrate the Spectral Distance Error (SDE) with mean and standard deviation across directions and subjects in the top and bottom plots, respectively. As this research utilizes different data, it is not directly comparable with other studies. Because the used data is half synthetic, which is likely easier to model than real data, this research thus provides a lower bound of what the SDE for HRTFs might have been with use of such ML models.

Another interesting paper titled "On the Relevance of the Differences Between HRTF

Measurement Setups for Machine Learning” [22] by J. Pauwels and L. Picinali focuses on the challenges associated with datasets aspect of using machine learning techniques for processing HRTFs. More precisely, the paper addresses the issue of limited size in existing HRTF datasets and the challenge of combining datasets from various sources. Reserchers standardised existing datasets, bringing them to a common ground by ensuring they all have the same sample rate, impulse length, etc. Then, they trained classifiers to distinguish between different datasets. With different classifiers and ranges of frequencies retained in the HRTFs, the accuracy ranged from as high as 95% to as low as 45% (still higher than a random guess). This suggests that every dataset includes unique spectral information serving as a signature, enabling the classifier model to identify the institution or lab from which the dataset was collected. The researchers further tested the classification strength by removing frequencies from the HRTFs and discovered that the lowest and highest frequencies contribute the most to classification accuracy. They hypothesized that the low frequency might contain a room response form the lab, while the high frequency might be characteristic of the equipment. The implications of these findings suggest that any machine learning workflow involving HRTFs is at risk of generalizing poorly to other measurement setups. This means that simply mixing datasets to increase the number of training examples will not automatically enhance robustness. Additionally, cross-dataset testing of the results is crucial.

1.5 Experiments, Results and Analysis

In this section, we present several experiments focusing on HRTF personalization efforts and are presenting the outcomes herein. Our methodology aligns with and extends the work detailed in "Head Related Impulse Response Interpolation and Extrapolation Using Deep Belief Networks," [13] authored by G. Kestler, Shahrokh Yadegari, and D. Nahamoo, published in 2019. This study, discussed previously (see Section 1.4.2), introduced a method employing a stack of fully connected neural networks for the interpolation and extrapolation of HRTF databases using existing HRTF data alongside anthropometric measurements of CIPIC subjects. Our investigation explored the potential of advancing this method to generate HRTFs for any spatial position around the head, solely based on an individual's anthropometric data.

Throughout the research and adaptation of the method proposed in the past study to personalisation task, we encountered an overfitting problem due to small size and insufficiency of the CIPIC data. We varied training parameters such as the learning rate and its scheduling and we experimented with different amounts of dropouts. We also tested the effects of masking and weighting a loss function to guide the model's focus towards certain spatial areas. We explored and tested the efficiency of the original paper's method, which involves intermediate steps and stacking trained networks. We also tried to clean and correct the CIPIC dataset, which lacks certain anthropometric data, and to mitigate error artifacts we think are result of recordings process inefficiencies. We explored the possibilities of improving model metrics through this approach by training a model on the adjusted data.

The subsequent sections detail our experiments, providing quantitative results and visual representations, especially focusing on LSD numbers and graphs, together with synthesized HRTFs for some meaningful data points across azimuth and elevation for both training and testing subjects. Our report will highlight findings that either showed some meaningful results or offered insightful revelations regarding HRTF generation challenge.

1.5.1 Base Model

The first experiment we conducted involved training existing models from [13] on full CIPIC dataset without excluding any positions, allowing the network to learn the full space of the data for each subject. We utilized all 35 subjects with available anthropometric data from CIPIC and divided the data into training, validation, and test sets with proportions of 70%, 20%, and 10% respectively (resulting in 4 test subjects). We

trained all networks for 20 epochs and 20 it-

erations, mixing validation data and removing new subjects as validation data as described in the original paper, except for the "magl" and "magr" networks, which we trained for 25 epochs and 25 iterations. We then evaluated whether the network could successfully predict HRTF curves for test subjects it had never seen in the time of training, given anthropometric data and a set of positions. The learning rate for the experiment was set to 0.0005.

The Table 1.5.1 shows LSD numbers for the experiment, for both left and right ears across two different frequency ranges: the full spectrum and below 11kHz.

The Figure 1.5.1 shows LSD graphs for training subject 003 and one of the test subject 009. It is evident that the results for the

test subject are not favorable, while the training subject shows a overly good approximation. This is a clear indication of overfitting, which is expected for such a small dataset as CIPIC.

To provide further insight into the model's performance, Figure 1.5.2 showcases recon-

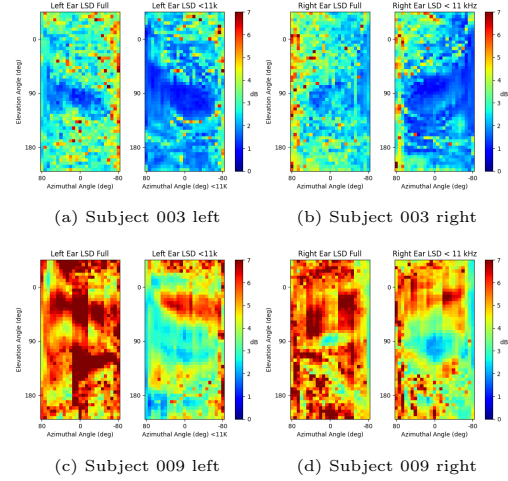


Figure 1.5.1: HRTF LSD images for all positions for left and right ear subject 003 from training set and 009 from test set.

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.844	3.983
	Right	5.489	4.402
Training	Left	3.138	2.275
	Right	3.034	2.353
Test zero elevation	Left	5.238	4.979
	Right	5.332	3.956

Table 1.5.1: LSD Results for base model

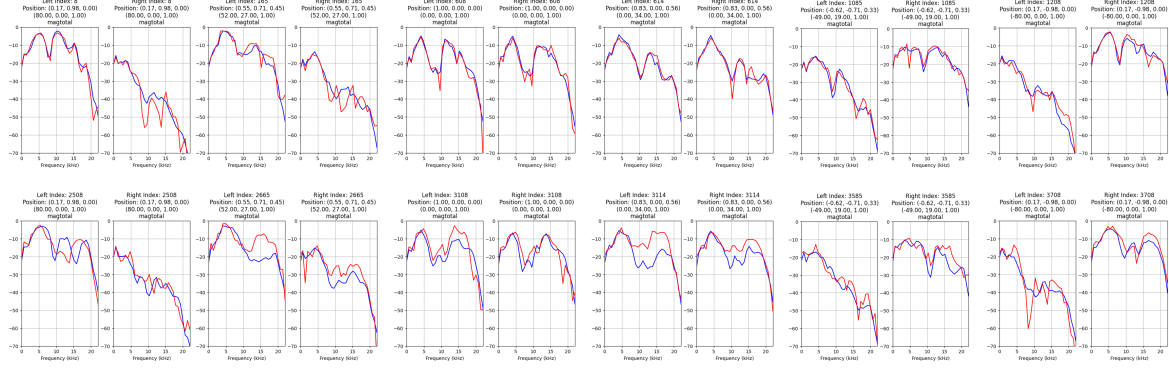


Figure 1.5.2: HRTF LSD images for some important positions for the left and right ears of subjects 003 from the training set (upper row) and 009 from the test set (lower row).

structions for several crucial data points. From left to right, the depicted points include the leftmost position at azimuth zero, the left halfway point with elevation, the center with no elevation, and the center with elevation. Similarly, reconstructions for the corresponding points on the right side are also displayed. These points are presented for training subjects 003 on the upper row and for the test subject 009 on the lower row. While reconstructions for both subjects show relatively meaningful matches to real HRTF curves, those for the test subject often exhibit mismatches and training shows too precise approximation, again indicating a potential overfitting problem. This discrepancy highlights the importance of robust model evaluation and the need for techniques to mitigate overfitting in our models.

1.5.2 Dropouts

Continuing our research, we tried dropout regularization as a remedy to the overfitting problem. By introducing dropout regularization, we aim to improve the generalization capabilities of our model and reduce overfitting tendencies. Accordingly, we attempted to address this issue by implementing various dropout levels into our training process. We kept all the parameters of the model, data, and training procedure constant, same as in the baseline. First, we experimented with dropout rate of 10%. The results of the experiments are summarized in the table table 1.5.3 and corresponding LSD figures are shown in the figure 1.5.4.

As illustrated by the data presented in the table and figure, the model exhibits a marginal improvement in the test LSD and slightly worse results for the training subjects, which might be interpreted as a reduction in overfitting. However, upon closer examination, most generated HRTFs appear similar across data points, and the reconstruction reveals that the model is clearly underfitting the data. In Figure 1.5.5, the predictions for both the test and training subjects, as well as between different points, predominantly reflect the mere averages of the curves, offering only general trends. Such outcomes cannot be deemed satisfactory for our research objectives. This issue mirrors the problems reported in the work of Miccini et al. [21], highlighting a recurring challenge in accurately modeling HRTFs.

We also experimented with various dropout rates, encountering issues similar to those described above. Although these adjustments led to numerical improvements, the resulting reconstructions lacked the necessary detail for accurate approximation of the HRTF function, which is crucial for our purposes. For brevity, we will not present de-

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.325	3.857
	Right	5.128	4.000
Training	Left	4.788	3.925
	Right	4.724	3.914

Figure 1.5.3: LSD Results for dropout 10% model

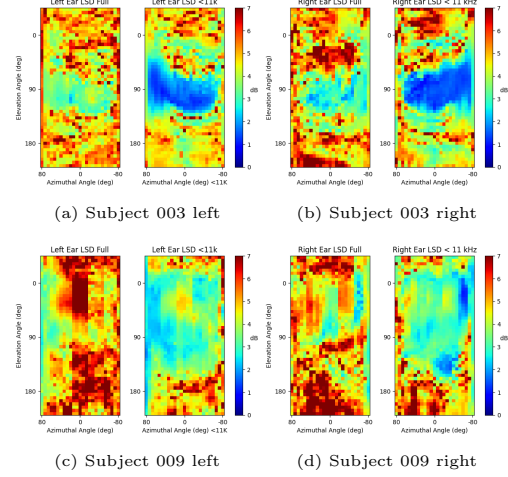


Figure 1.5.4: HRTF LSD images for subject 003 and 009 for the model with dropout rate 10% .

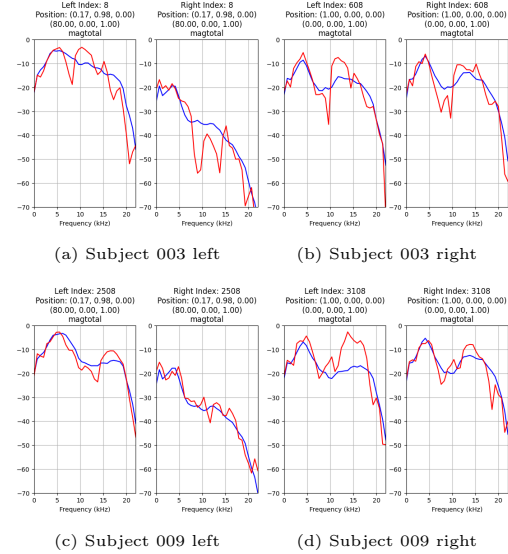


Figure 1.5.5: HRTF curves for a few positions for subject 003 (training) and 009 (test) for the model with dropout rate 10%.

tailed figures for these results; instead, we will report only the LSD numbers for test subject 009 in the Table 1.5.2.

Dropout Rate	Subject	Ear	LSD full	LSD <11kHz
1%	Test 009	Left	5.971	4.107
		Right	4.987	3.766
5%	Test 009	Left	5.393	3.575
		Right	5.306	3.544
20%	Test 009	Left	5.500	3.94
		Right	5.521	4.11

Table 1.5.2: LSD Results for Various Dropout Levels

1.5.3 Learning Rate

Another avenue of our experimentation was focused on adjusting the learning rate, incorporating its scheduling, and implementing early stopping. We employed a scheduler that reduced the learning rate by half whenever there was no improvement in validation loss. Additionally, we experimented with both small and large learning rates. For the sake brevity, we will not provide detailed figures from these experiments, as we believe they did not yield any significant, valuable improvements or new insights into solving the problem.

The results of these experiments are detailed in Table 1.5.3. As indicated, we practically did not benefit from implementing learning rate scheduling, and introducing a higher learning rate of $1e-3$ resulted in even more overfitting compared to $5e-4$, the rate used for our base model. We consider a learning rate of $5e-4$ to be optimal, and we believe the Adam optimizer, which we utilized for training, adequately handles basic scheduling and necessary adjustments. Given that we are not dealing with complicated

Learnng Rate	Subject	Ear	LSD Full	LSD <11kHz
lr-scheduler	Test	Left	5.782	4.253
		Right	5.025	3.961
	Training	Left	4.644	3.625
		Right	4.417	3.425
lr=1e-3	Test	Left	6.213	4.165
		Right	6.046	4.92
	Training	Left	2.713	2.049
		Right	3.201	2.482

Table 1.5.3: LSD Results for Various Dropout Levels

models or fine-tuning, the Adam optimizer effectively manages the learning rate for our model.

1.5.4 Remove Mean

We normalized the HRTF data by removing the mean calculated on training data from every frequency bin, aiming to simplify the model’s task by working with a smaller range normalized data. This approach was inspired by the method introduced in [9]. However, we found that removing the mean did not benefit the model. The results are presented in Table 1.5.4, which details the numbers associated with this experiment.

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.552	3.896
	Right	6.098	5.434
Training	Left	3.053	2.270
	Right	2.769	2.086

Table 1.5.4: LSD Results for Mean Remove

1.5.5 One Model

The original paper (see Section 1.4.2) presented a methodology involving intermediate steps of model training for different aspects of the final HRTFs, followed by stacking trained networks before training the final model. In line with our focus on fully connected networks, we tested the hypothesis of the efficiency of this method against training only one fully connected network for matching anthropometric data to HRTFs. Table 1.5.5 presents the results of the above experiment, while Figure 1.5.7 displays the LSD images for frequencies. As before, we show numbers and graphs for subject 003 from the training dataset against subject 009 from the test dataset.

Subject	Ear	Full Spectrum	Below 11kHz
Test	Left	5.837	3.652
	Right	5.863	4.012
Training	Left	3.556	2.746
	Right	3.567	2.702

Table 1.5.5: LSD Results for One Model

As we can observe, there is a small improvement for LSD values for frequencies below 11kHz with one model paradigm compared to base model, which is clearly observed in Figure 1.5.7. We investigated the possibility of "average curve" problem that we encountered case of dropout, but upon examining the individual HRTF plots, it is evident that although the reconstruction is not perfect, the curves do not lack detail. We conclude that the two methods used in the Base model and One model are roughly equivalent in their performance. The only difference lies in efficiency; the proposed method from the original paper takes more than 12 hours to train as it trains 12 different models in a row stacking them to each other to construct final model, while the One Model paradigm takes around 1 hour. Some subsequent experiments were

conducted with the One Model approach aim to save time and resources, which will be presented in the following sections.

1.5.6 Masking and Weighting

In the series of following experiments, we aimed to test the hypothesis that the proposed network, or fully connected networks generally in this regard, could address the given problem by focusing on a smaller subset of the problem. This was achieved by limiting data position

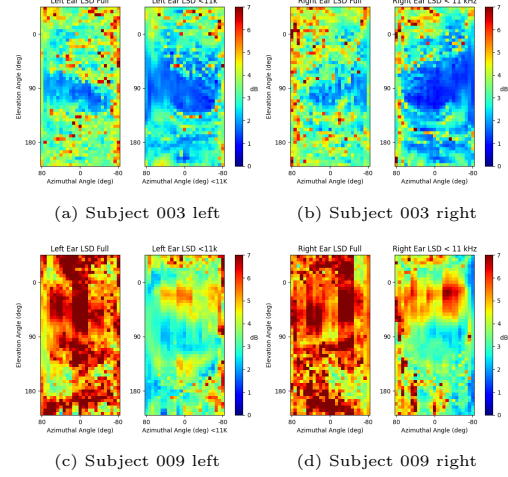


Figure 1.5.6: HRTF LSD images for subject 003 and 009 for the One Model.

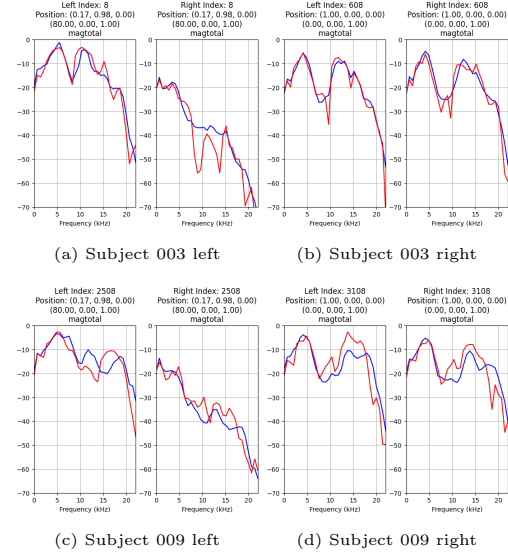


Figure 1.5.7: HRTF curves for a few positions for subject 003 (training) and 009 (test) for One Model.

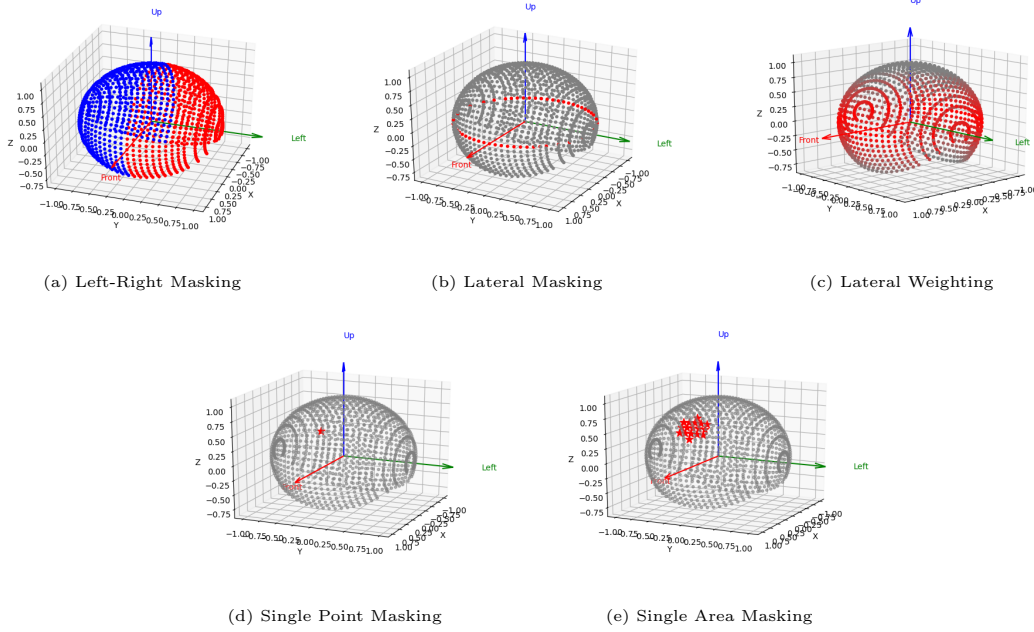


Figure 1.5.8: Different variations of masking and weighting methods for CIPIC data positions.

in the dataset or directing the model’s attention to a subset by applying weighting or the masking to loss function of the model during training. We conducted several experiments that explored such methodologies:

Lateral Masking: In this experiment, we masked all data points in the CIPIC dataset except for those at elevation zero. The goal was to assess whether our model could learn HRTF reconstruction in 2D space. Graphically, this masking procedure is shown in Figure 1.5.8b, where positions in red represent the only positions the model was trained on. Only positions in red were given loss weight of 1, and all other positions were given a weight of 0.

Lateral Weighting: This experiment serves as a more generalized version of the previous one. Instead of training the model solely on a narrow stripe at the no-elevation position, we assigned a weight to the loss function that is 1 at elevation zero and decreases inversely proportionally to the elevation, reaching 0 at the highest elevation point of the CIPIC data sphere. This approach is depicted in Figure 1.5.8c, where the intensity of the red corresponds to the weight given to the loss function.

Left-Right Masking: This experiment divides the CIPIC sphere into two halves, and

the model’s loss function is masked accordingly. The left outputs of the model calculate their loss considering only left positions, and the right outputs consider only the right side of the CIPIC sphere. This setup is illustrated in Figure 1.5.8a, where red and blue correspond to the left and right loss weights, respectively. This experiment tried to test if the models can divide their leaning capacity in two and learn data in such way.

Left-Right Weighting: This experiment is a generalized version of the previous one, where, instead of abruptly masking, we assign the most leftward position a loss weight of 1 for the left output of the network and decrease it linearly to the right, approximating 0 at the rightmost position. The same is done but vice versa for the right output of the model. This experiment aimed to test whether the model can focus the attention of its outputs to the corresponding sides and thereby improve generation quality. Additionally, we conducted a reversed version of this experiment where the left output of the model has a loss weight of 1 on the right side of the CIPIC sphere. This was done to evaluate if the model can focus its attention on the harder tasks of leaning farther positions and to explore the possibility of merging the solutions of both models if they prove successful.

Single Point Masking: This experiment tests the minimal hypothesis of whether the model can learn at least one position for all subjects and precisely generate a new point for test subjects given only anthropometric data, with position data fixed for all subjects. This concept is depicted in Figure 1.5.8d, where only the front upper position in front of the human subject receives a weight of 1 and actually participates in loss calculation and model training. This experiment effectively reduces the practical data size to 35 entries. We tested this approach with different positions in space, including left, right, and center at non-zero elevations.

Single Area Masking: This experiment represents an expanded version of its predecessor, concentrating the model’s learning efforts on a specific sector of the CIPIC sphere. The objective was to evaluate the model’s proficiency in mastering a discrete segment of the task, with the broader goal of employing multiple models in tandem should this strategy

prove successful. The design of one of the examples of this experiment is illustrated in Figure 1.5.8e, where only a small area centered in front and above the human subject is selected with a tolerance of 0.2 from the center. Only positions with a vector distance of less than 0.2 from the center target point are assigned a loss weight of 1, while all other positions are excluded from training by setting their loss to 0.

For the sake of convenience, We report the results of the aforementioned experiments by displaying LSD numbers and LSD images solely for the test subject 009. Table 1.5.6 and Figure 1.5.9 showcase these results. The inclusion of LSD images is particularly advantageous as full LSD numbers may not provide informative insights for models trained only on specific areas of points. Therefore, focusing on LSD values within the trained and attended regions is crucial for a comprehensive understanding.

To interpret the results of the experiments, let us examine Table 1.5.6 and Figure 1.5.9 systematically from top to bottom and left to right.

In the first experiment, Lateral Masking, despite being trained on only 50 data points (out of a total of 1250 positions), the model shows promising results. However, it falls short of its primary objective to maintain a low LSD score along the zero-elevation surface. This shortcoming is evident in the LSD images, where, contrary to the anticipated green or blue areas around zero elevation horizontal stripe indicating lower error, we observe red areas denoting higher error in both the left and right ears images. This discrepancy underscores the model’s inability to sustain an LSD below 4 dB, even for the simplified task of focusing

Experiment	Subject	Ear	LSD full	LSD <11kHz
Lateral Masking	Test 009	Left	5.223	4.298
		Right	5.479	4.669
Lateral Weighting	Test 009	Left	6.077	4.367
		Right	5.583	4.114
Left-Right Masking	Test 009	Left	6.945	4.828
		Right	6.348	4.166
Left-Right Weighting	Test 009	Left	6.016	4.356
		Right	5.349	3.734
Left-Right Weighting Reversed	Test 009	Left	5.813	4.016
		Right	5.161	4.089
Single Point Masking ($\theta = 52, \phi = 63$)	Test 009	Left	10.988 ¹	9.150 ¹
		Right	12.693 ¹	9.827 ¹
Single Area Masking ($\theta = 52, \phi = 63, \tau = 0.2$)	Test 009	Left	8.722 ¹	7.275 ¹
		Right	12.452 ¹	9.472 ¹

Table 1.5.6: LSD Results for Various Masking and Weighting Strategies

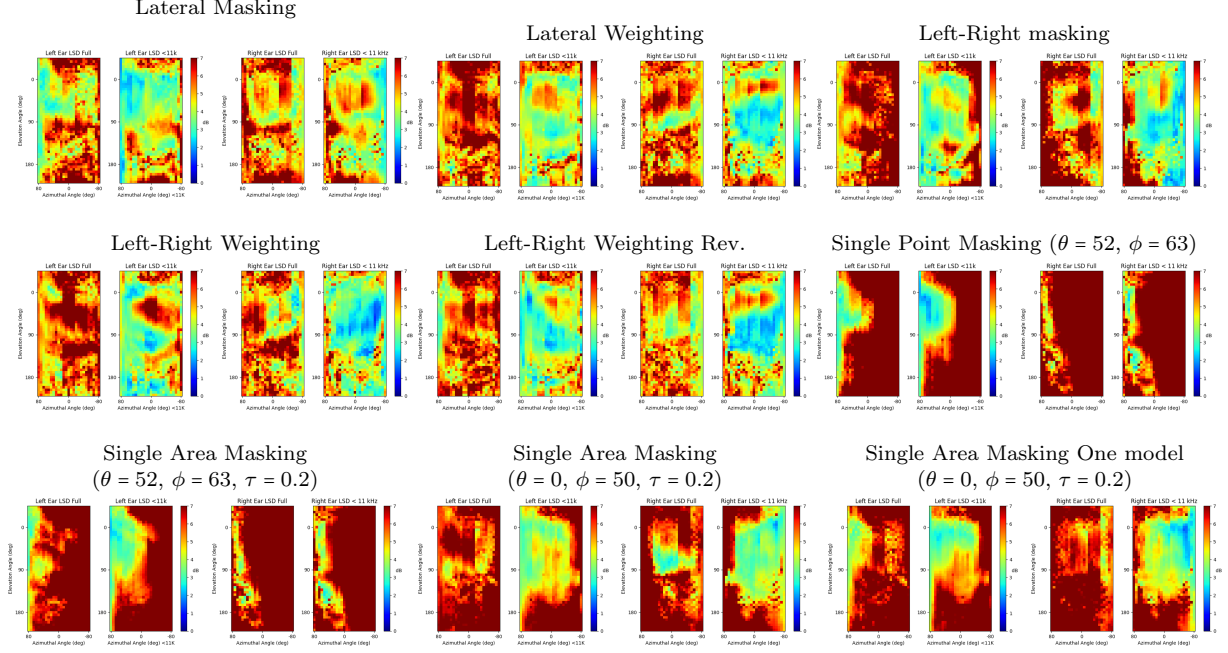


Figure 1.5.9: LSD Images for Various Masking and Weighting Strategies

on the no-elevation positions it was trained on.

Proceeding further, we encounter the Lateral Weighting experiment, which proved to be less effective than the baseline model, not only in reconstructing the zero-elevation stripe but also positions above and below it. Intriguingly, it is evident that the previous model, which focused exclusively on zero-elevation positions, predicted some areas in front and above the human subject (0-degree to 90-degree area in the image) more accurately than this one.

Next, we examine the Left-Right Masking, which also proved to be substantially inefficient. Essentially, the model fails to predict HRTFs for the positions opposite to given ear effectively, which aligns with the limitations imposed by the Masking constraint. However, surprisingly, the model does not show improvement even in the areas where its focus was directed, still exhibiting numerous red areas, indicating high errors even there.

The Left-Right Weighting and Left-Right Weighting Reversed exhibited significantly better results than the other experiments in this set and almost mirrored the numbers and images produced by the base model. However, we did not observe the anticipated outcome where Left-Right Weighting would display more green and blue areas, indicating superior

reconstruction in the azimuth space from 80 to 0 degrees on the left and 0 to -80 degrees on the right side. Conversely, Left-Right Weighting Reversed was expected to exhibit the effect in the opposite direction. This could have allowed us to divide the task of HRTF estimation between two parallel models, predicting the left ear HRTF from the model that assigned more weight to the corresponding side of ear position, and predicting the opposite ear HRTF from a model that placed more weight on the opposite side. This outcome suggests that the model does not benefit from decomposing the task, pointing to a more fundamental issue. We hypothesize that this could be related to the quantity and significance of the data used for prediction, as well as the meaningfulness of the parameters for data collection, indicating deeper underlying problems beyond the design of the ML model itself.

To further explore the task of breaking down the problem into smaller sub tasks and testing the model’s capacity to solve smaller, partial problems—which could later aggregate into insights for a more comprehensive solution—we evaluated the model’s ability to accurately predict a single point. The selected point was position index 165, corresponding to a left side azimuth of 52 degrees and an elevation angle of 63 degrees. While the reconstruction for the left ear was not perfect, it was somewhat tolerable; however, the model entirely failed to accurately predict for the right ear’s HRTF. Additionally, an interesting observation from the single-point experiments was that, regrettably, the model does not differentiate between subjects, predicting an average curve that seems to minimize loss effectively for the small dataset we are getting after focusing on one point. This phenomenon might be attributed to so-called degenerate solutions problem in ML, where the model converges to a trivial solution. We believe this could be a result of the dataset being either too small or not representative of the underlying data distribution, leading the model to fail in learning meaningful patterns and instead resort to generating constant predictions that satisfies the local minima of loss function. In this case, the dataset size is 31 items in training+validation set, with input sizes of 17+10 and an output size of 64. We suspect this might be a reason for the problem.

To address the problem of an overly small dataset and to render the model more meaningful, we experimented with directing the model’s full attention to a small area of data point positions instead of one point. This approach seemed logical not only because of intuitions from physics and the general perception that sound from areas closer to each other tends to be similar but also because it was substantiated by the original paper and numerous other studies. These sources indicate there is always a strong correlation between neighboring positions in the dataset, between which interpolation and extrapolation algorithms have performed well. We hypothesized that a group of data points would provide the model with enough information for effective training, enabling it to learn the defining forces in anthropometric data. We took the same data point of index 165, and instead of focusing on a single point, we introduced a tolerance area with $\tau = 0.2$, where every position within a vector distance less than 0.2 from the center point would be included in the area. Unfortunately, the model again proved inefficient, failing to reconstruct the HRTF data for the opposing ear accurately. To further test this hypothesis, we additionally explored the model’s performance on several different positions and observed that the model does not efficiently learn HRTF curves. We present data point 615, which corresponds to a mid-front position with an elevation angle of 50 degrees relative to the human subject. As observed in the images, the model fails to accurately predict this area for the left ear while performing relatively well for the right ear.

Lastly, for the sake of experiment, we trained the One Model variant on the same single area task, to determine its capability to learn the front upper positional area (position index 615). As illustrated in the last image on the bottom right side of Figure 1.5.9, the model exhibits no improvement over the base model that was focused on the same area.

In conclusion, our series of experiments on HRTF prediction models revealed limitations in accuracy across various approaches, including lateral masking and weighting, and focused attention on specific data points or areas. There were no improvements and Model did not behave in the way we imagined they would. These findings suggest the existence of some

fundamental problems in the task of predicting HRTF data from anthropometrics, at least with CIPIC dataset and the given setting.

1.5.7 Cleaned and Smoothed CIPIC

The CIPIC dataset was found to have several problems by different generations of researchers at UC San Diego while conducting research and working on [13] and its continuation, of which this chapter is also a part. Several problems were identified, such as inconsistencies in HRTF measurements, unexpected jumps in some frequency bins, and most importantly, the "5th ring problem," which showed systematic inconsistencies in

Experiment	Subject	Ear	LSD full	LSD <11kHz
notch-smoothing-1	Test 009	Left	5.616	3.745
		Right	5.421	3.987
notch-smoothing-2	Test 009	Left	5.111	3.652
		Right	5.511	3.884
notch-smoothing-3	Test 009	Left	5.387	3.834
		Right	5.579	4.226
5thring-6	Test 009	Left	5.405	3.854
		Right	5.243	4.147

Table 1.5.7: LSD Results for Various Masking and Weighting Strategies

the 5th ring of the CIPIC spherical ring. This issue was attributed to the problem of a changed volume of the speaker responsible for the 5th ring. These problems were addressed, and the CIPIC data was cleaned and smoothed in several ways, producing four different versions of the CIPIC data: "notch-smoothing-1," "notch-smoothing-2," "notch-smoothing-3," and "5thring-6," with the latter specifically addressing the "5th ring problem." We will not go through the details of smoothing and cleaning the CIPIC data as this was not done by the author of this chapter, and there is a lack of good documentation on the exact procedures.

We conducted experiments and tried to train our model with all the types of cleaned and smoothed CIPIC data. The results of the experiments are given in Table 1.5.7. Unfortunately, as observed in the table, the cleaned and smoothed CIPIC data, in any of its iterations, does not provide any substantial improvement to the model. This suggests that the problem is not simply the cleanliness or smoothness of the data but may be more fundamental to the ML modeling of CIPIC data.

1.6 Conclusion

In this chapter, we explored the theory and practice behind spatial sound, focusing on Head-Related Transfer Functions (HRTFs) and their modeling and personalization using ML. We reviewed the theoretical background of HRTFs and discussed the challenges in the spatial audio field, particularly regarding HRTF acquisition and synthesis using ML-based methods. We detailed the measurement process and the challenges of acquiring accurate HRTF data, which have driven the development of ML models in this field. We presented existing datasets and methods for evaluating simulated HRTFs. Following this, we reviewed existing literature and methodologies for HRTF generation with ML methods. Finally, we detailed our attempts to tackle the problem of personalizing HRTFs with ML through various methodologies, addressing the challenges of overfitting and generalization in small datasets like CIPIC.

Our exploration of the literature and our own experiments revealed that, despite tremendous amount of work and effort directed to the field, the problem of generating HRTFs from anthropometric data still remains largely unresolved. We believe, the key issues of the field lie in the lack of extensive datasets and the lack of precision of existing measurement processes. We found that most existing work, especially ones working with current datasets, do not produce satisfactory results, even when using state-of-the-art ML models and methods. This indicates that current datasets, like CIPIC, are not sufficient for training robust ML models for HRTF personalization. Similarly, despite various approaches we tried, our model failed to generalize effectively and generate satisfactory results. To our qualified satisfaction, we observed that our results fall within the same range of LSD error as those from other more recent and sophisticated methods. While it is reassuring that our approach is somewhat reasonable, it is also disappointing as it once again proves that the problem is generally unresolved. Moreover, our exploration suggests that beyond dataset limitations, there might be fundamental aspects of HRTFs that are not yet fully understood, preventing ML methodologies from modeling them correctly. The absence of standardized methods for

measurements, the lack of a precise list of necessary anthropometric data, and other factors possibly add to the complexity of HRTF modeling. These insights point towards the need for more comprehensive datasets, better understanding of the theory, and standardized measurement processes, which could form the basis for future research in this challenging but exciting field.

Bibliography

- [1] V. Ralph Algazi, Richard O. Duda, Dennis M. Thompson, and Carlos Avendaño. The cipic hrtf database. Proceedings of the 2001 IEEE Workshop on the Applications of Signal Processing to Audio and Acoustics (Cat. No.01TH8575), pages 99–102, 2001.
- [2] Sunil G. Bharitkar, Timothy Mauer, Teresa Wells, and David Berfanger. Subspace-based hrtf synthesis from sparse data: A joint pca and ml-based approach. In Audio Engineering Society Convention 145, Oct 2018.
- [3] Piotr Bilinski, Jens Ahrens, Mark R. P. Thomas, Ivan J. Tashev, and John C. Platt. Hrtf magnitude synthesis via sparse representation of anthropometric features. In 2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 4468–4472, 2014.
- [4] Fabian Brinkmann, Manoj Dinakaran, Robert Pelzer, Peter Grosche, Daniel Voss, and Stefan Weinzierl. A cross-validated database of measured and simulated hrtfs including 3d head meshes and anthropometric features. Journal of the Audio Engineering Society, 2019.
- [5] Tzu-Yu Chen, Tzu-Hsuan Kuo, and Tai-Shih Chi. Autoencoding hrtfs for dnn based hrtf personalization using anthropometric features. In ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 271–275, 2019.
- [6] Chan Jun Chun, Jung Min Moon, Geon Woo Lee, Nam Kyun Kim, and Hong Kook Kim. Deep neural network based hrtf personalization using anthropometric measurements. In Audio Engineering Society Convention 143, Oct 2017.
- [7] Haytham Fayek, Laurens van der Maaten, Griffin Romigh, and Ravish Mehra. On data-driven approaches to head-related-transfer function personalization. Proceedings of the 143rd Audio Engineering Society (AES) Convention 2017, page 10, 2017.
- [8] Arthur E Hoerl and Robert W Kennard. Ridge regression: applications to nonorthogonal problems. Technometrics, 12(1):69–82, 1970.
- [9] Hongmei Hu, Lin Zhou, Hao Ma, and Zhenyang Wu. Hrtf personalization based on artificial neural network in individual virtual auditory space. Applied Acoustics, 69(2):163–172, 2008.

- [10] Ke Huang and Selin Aviyente. Sparse representation for signal classification. Advances in neural information processing systems, 19, 2006.
- [11] Ziran Jiang, Jinqiu Sang, Chengshi Zheng, Andong Li, and Xiaodong Li. Modeling individual head-related transfer functions from sparse measurements using a convolutional neural network. The Journal of the Acoustical Society of America, 153 1:248, 2023.
- [12] Brian F. G. Katz and Gaëtan Parseihian. Perceptually based head-related transfer function database optimization. The Journal of the Acoustical Society of America, 131(2):EL99–EL105, 01 2012.
- [13] Grady Kestler, Shahrokh Yadegari, and David Nahamoo. Head related impulse response interpolation and extrapolation using deep belief networks. In ICASSP 2019 - 2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 266–270, 2019.
- [14] Teuvo Kohonen. Self-organizing maps. Springs Series in Information Sciences, 30:362, 1995.
- [15] Sunil L Kukreja, Johan Löfberg, and Martin J Brenner. A least absolute shrinkage and selection operator (lasso) for nonlinear system identification. IFAC proceedings volumes, 39(1):814–819, 2006.
- [16] Geon Woo Lee and Hong Kook Kim. Personalized hrtf modeling based on deep neural network using anthropometric measurements and images of the ear. Applied Sciences, 8(11), 2018.
- [17] V. Lemaire, F. Clerot, S. Busson, R. Nicol, and V. Choqueuse. Individualized hrtfs from few measurements: a statistical learning approach. In Proceedings. 2005 IEEE International Joint Conference on Neural Networks, 2005., volume 4, pages 2041–2046 vol. 4, 2005.
- [18] Song Li and J. Peissig. Measurement of head-related transfer functions: A review. Applied Sciences, 2020.
- [19] Xiao li Zhong and Bo sun Xie. Head-related transfer functions and virtual auditory display. In Herve Glotin, editor, Soundscape Semiotics, chapter 6. IntechOpen, Rijeka, 2014.
- [20] Fei Ma, Thushara D. Abhayapala, Prasanga N. Samarasinghe, and Xingyu Chen. Spatial upsampling of head-related transfer functions using a physics-informed neural network, 2023.
- [21] Riccardo Miccini and Simone Spagnol. Hrtf individualization using deep learning. In 2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), pages 390–395, 2020.

- [22] Johan Pauwels and Lorenzo Picinali. On the relevance of the differences between hrtf measurement setups for machine learning. In ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 1–5, 2023.
- [23] Robert Pelzer, M. Dinakaran, F. Brinkmann, Steffen Lepa, Peter Grosche, and S. Weinzierl. Head-related transfer function recommendation based on perceptual similarities and anthropometric features. The Journal of the Acoustical Society of America, 148 6:3809, 2020.
- [24] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation, 2015.
- [25] Kanji Watanabe, Yukio Iwaya, Yôiti Suzuki, Shouichi Takane, and Sojun Sato. Dataset of head-related transfer functions measured with a circular loudspeaker array. Acoustical science and technology, 35(3):159–165, 2014.
- [26] Frederic L. Wightman and Doris J. Kistler. Monaural sound localization revisited. The Journal of the Acoustical Society of America, 101 2:1050–63, 1997.
- [27] R.S. Woodworth and H. Schlosberg. Experimental Psychology. Holt, 1954.
- [28] Kazuhiko Yamamoto and Takeo Igarashi. Fully perceptual-based 3d spatial sound individualization with an adaptive variational autoencoder. ACM Transactions on Graphics, 36:1–13, 11 2017.
- [29] Shu-Nung Yao, Tim Collins, and Chaoyun Liang. Head-related transfer function selection using neural networks. Archives of Acoustics, 42:365–373, 09 2017.
- [30] Xiang-Yang Zeng, Shu-Guang Wang, and Li-Ping Gao. A hybrid algorithm for selecting head-related transfer function based on similarity of anthropometric structures. Journal of Sound and Vibration, 329(19):4093–4106, 2010.
- [31] Bowen Zhi, Dmitry N. Zotkin, and Ramani Duraiswami. Towards fast and convenient end-to-end hrtf personalization. In ICASSP 2022 - 2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 441–445, 2022.
- [32] Yaxuan Zhou, Hao Jiang, and V. Ithapu. On the predictability of hrtfs from ear shapes using deep networks. ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pages 441–445, 2021.
- [33] Devansh Zurale, Shahrokh Yadegari, and Shlomo Dubnov. Deep hrtf encoding & interpolation: Exploring spatial correlations using convolutional neural networks. In 19th Sound and Music Computing Conference, SMC 2022, pages 350–357. Sound and Music Computing Network, 2022.